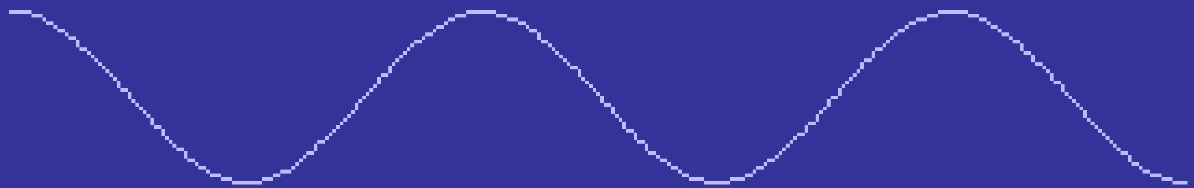
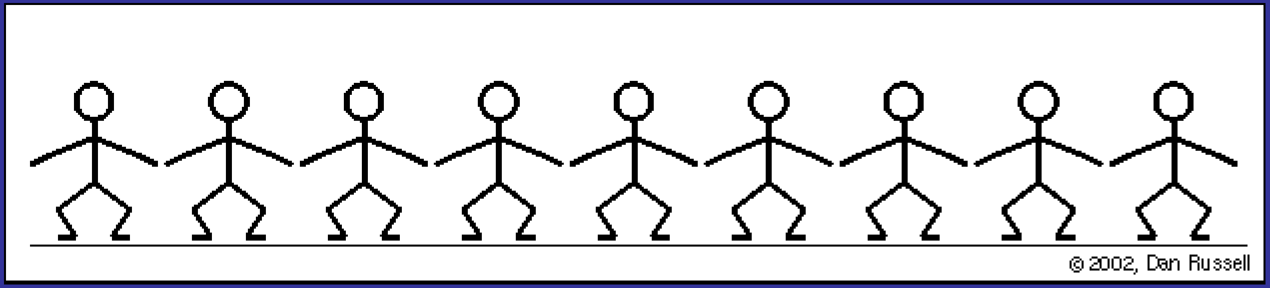
Waves





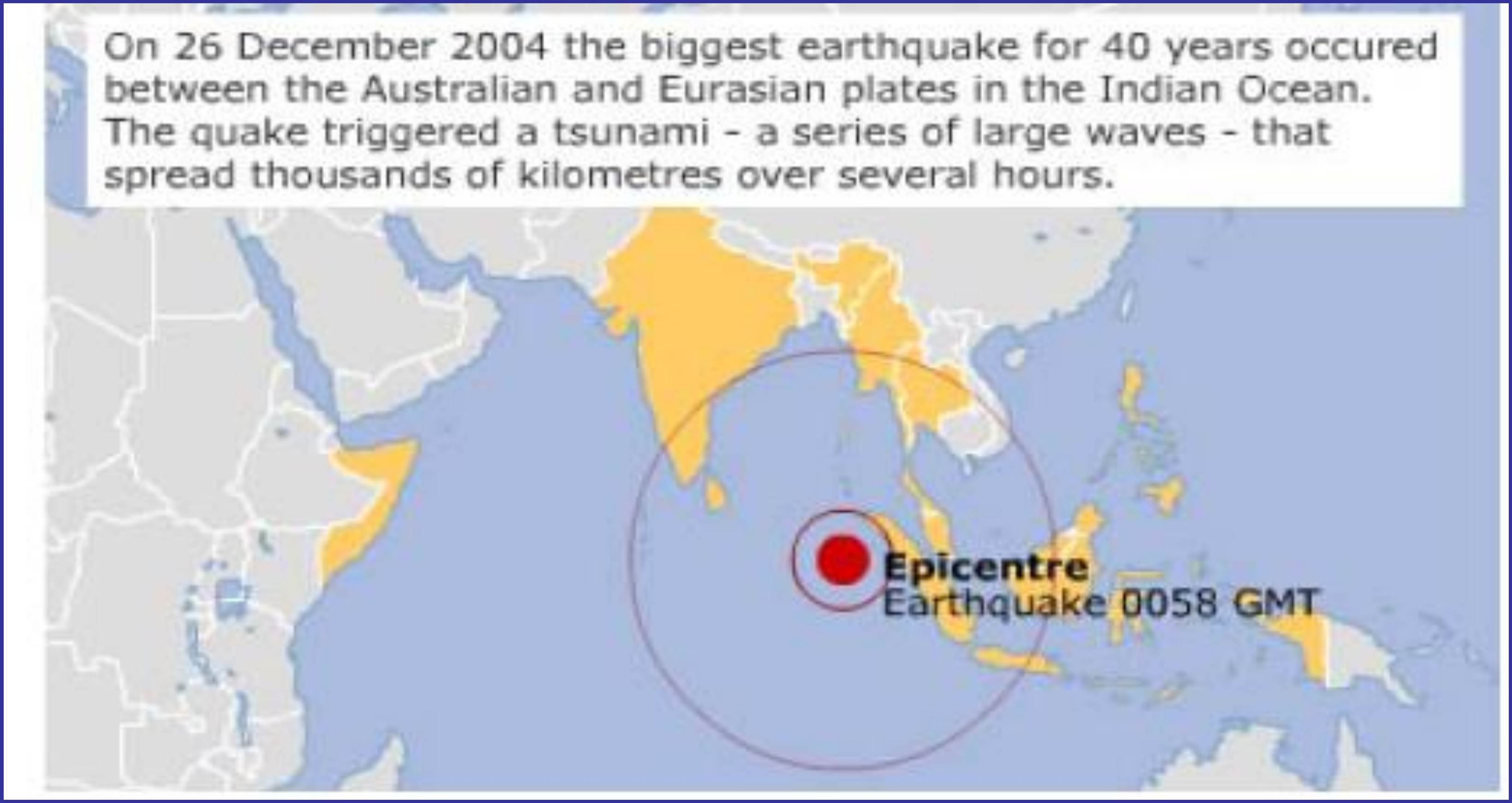
Wave – oscillations in time and space. Perturbation that propagates in space.

Type of waves:

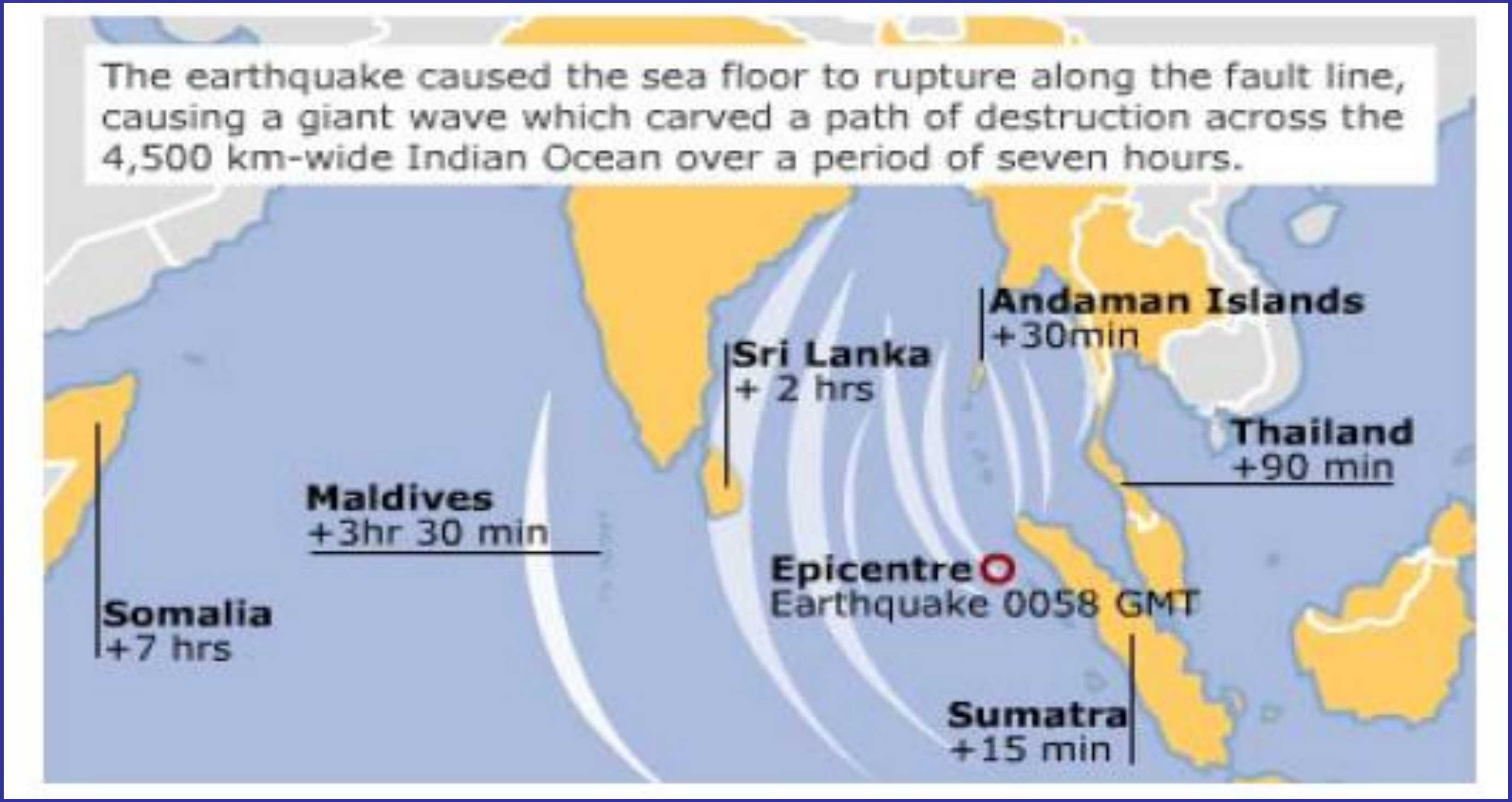
- mechanical (water waves, acoustic waves)
- electromagnetic waves (radio, microwaves, light)
- matter waves (electron as a wave?)

Wave carries energy and information

Transfer of energy by a wave



http://news.bbc.co.uk/1/hi/in_depth/4136289.stm



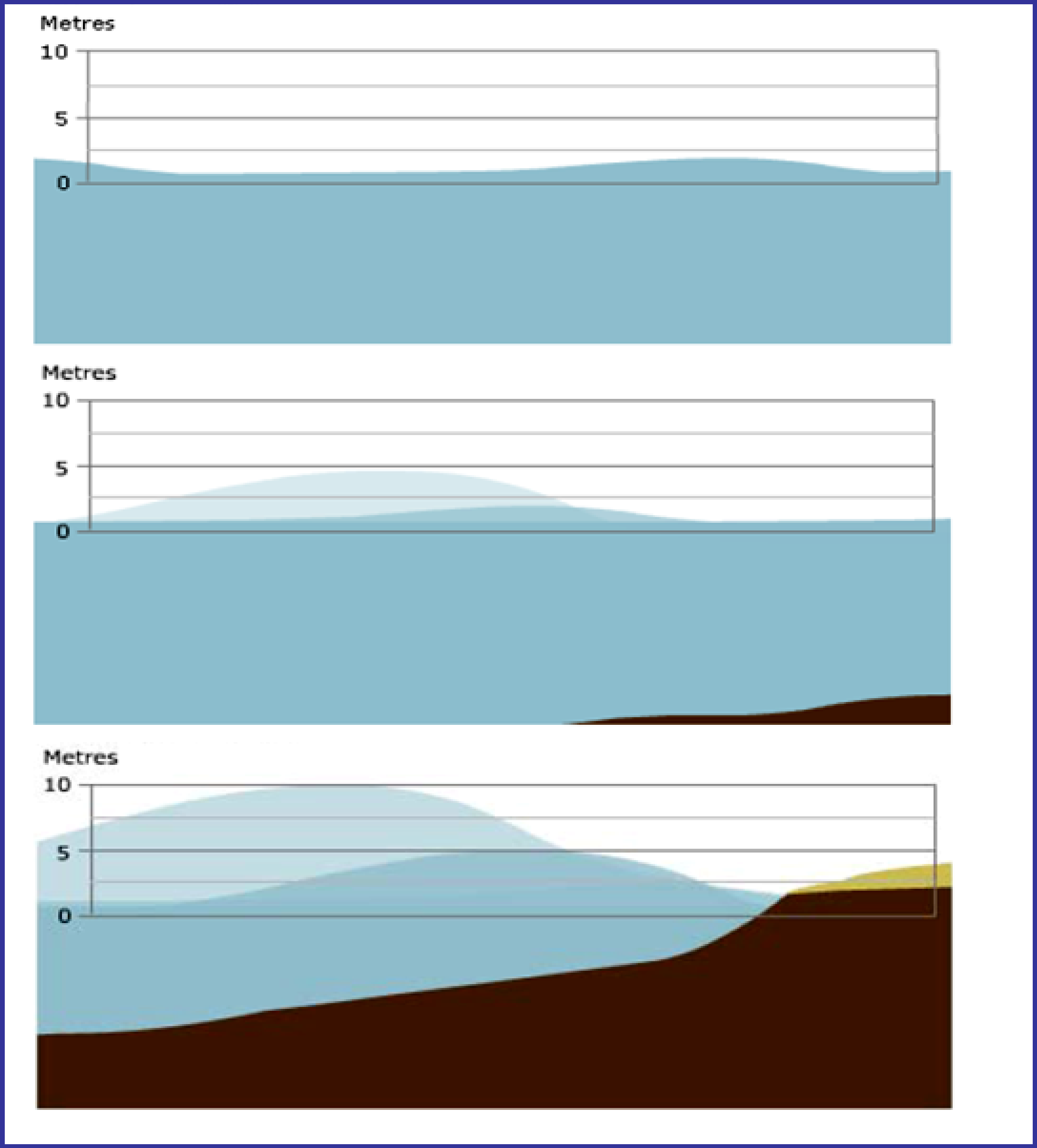
Tsoo-NAH-mee (jap.) waves

Tsunami waves on a deep water:

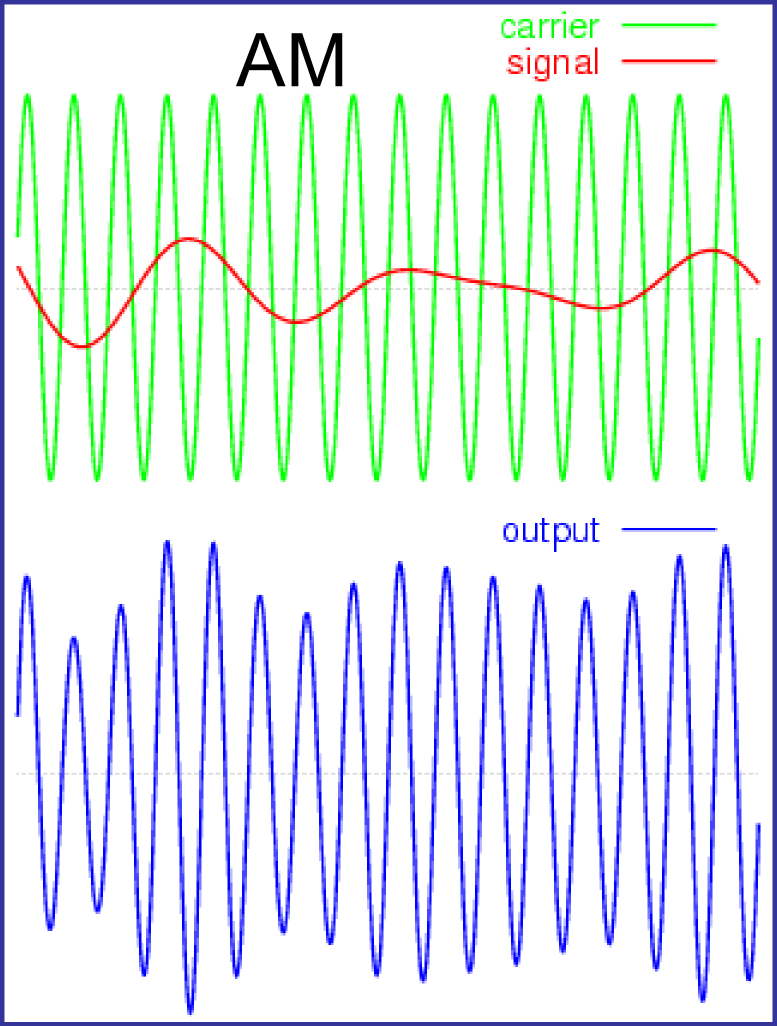
Small amplitude, big speed of propagation of 800 km/h

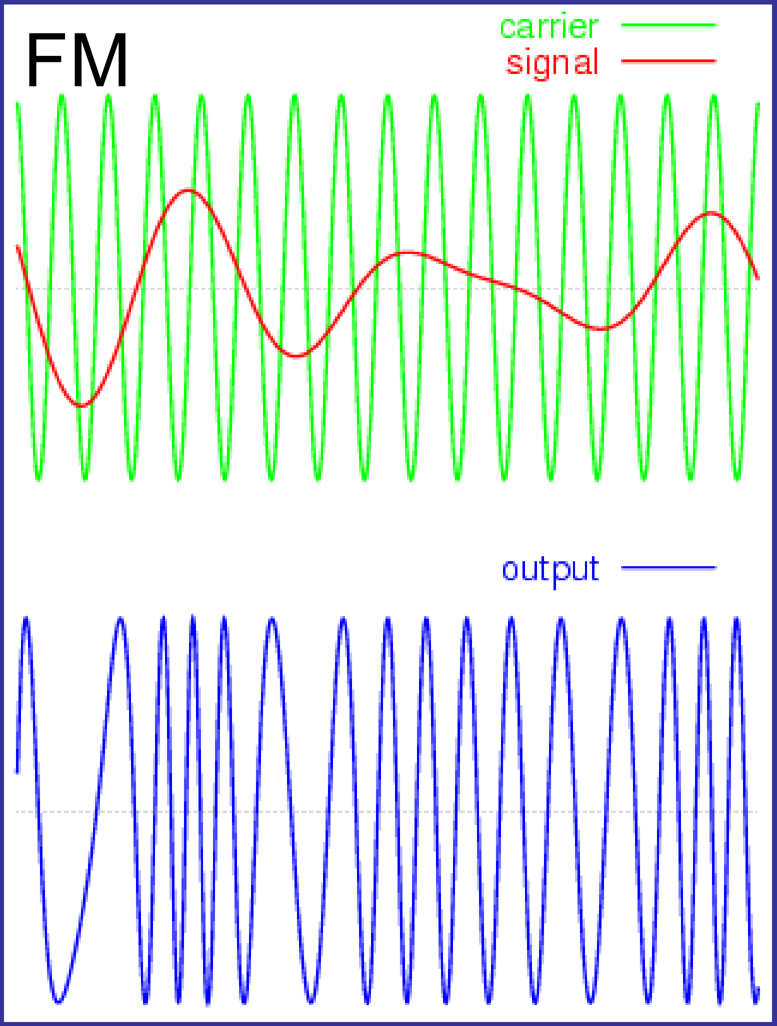
Tsunami waves on a shallow water:

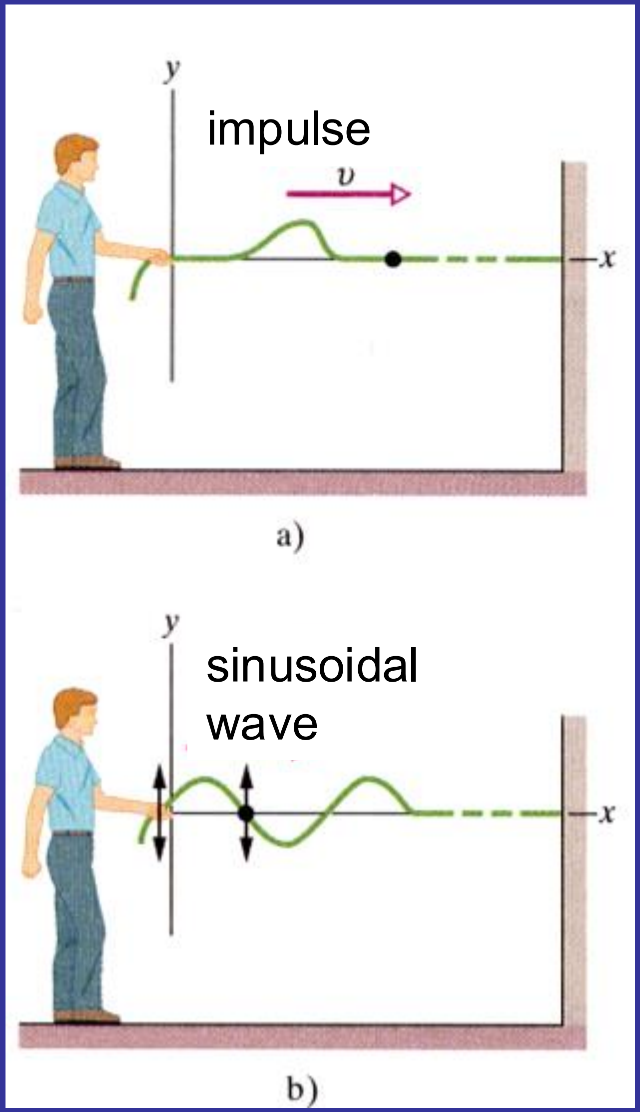
Smaller speed but much bigger amplitude (even up to 30 m)



Information? AM and/or FM modulation



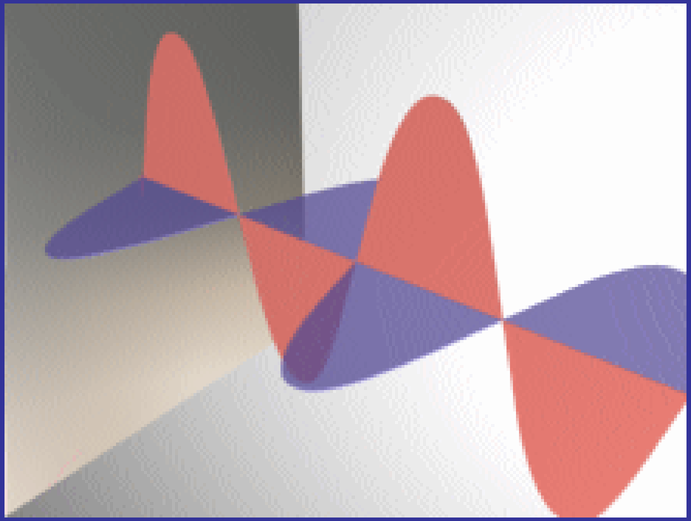




How to produce a wave?

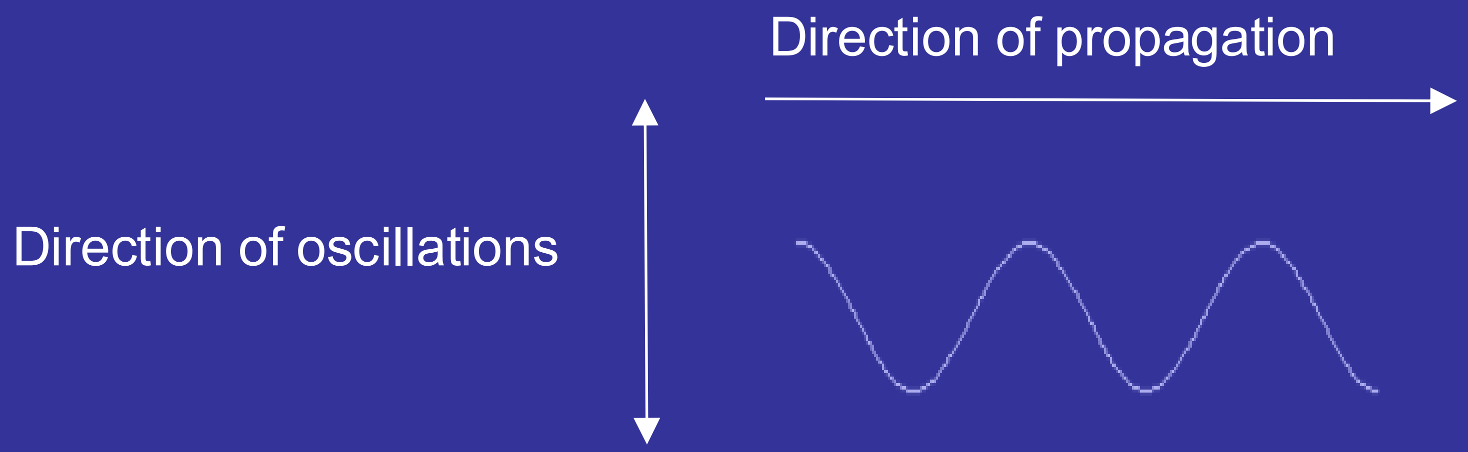
For mechanical waves propagating in a string, rod, air pipe (elastic medium), perturbation is a displacement from the equilibrium position, density, pressure. Wave arises when the element of the elastic medium is displaced from the equilibrium position. Material medium is necessary for propagation of the mechanical waves. Energy, not matter, is transported over a distance.

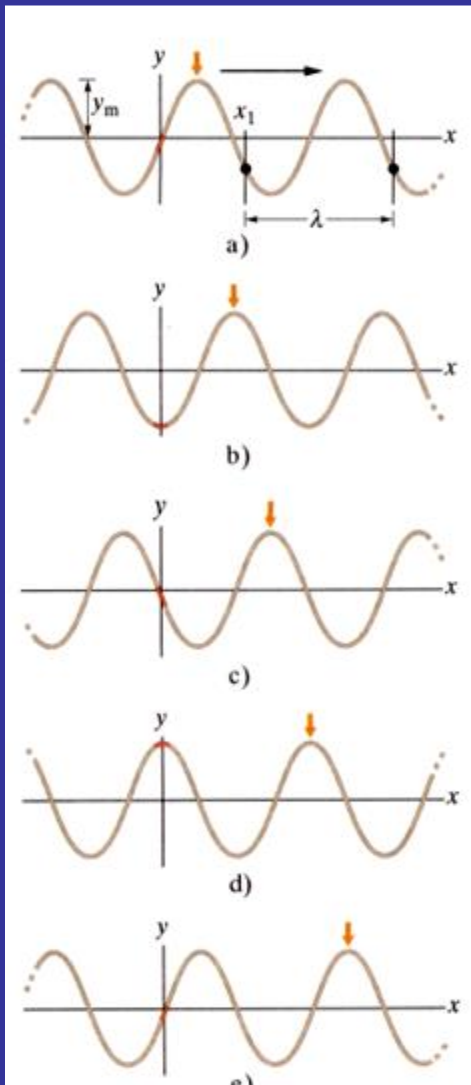
Electromagnetic wave (perturbation of electric E and magnetic B fields) propagates in vacuum – no medium is necessary



Experiment of Michelson and Morley, 1887 proves that „**Luminiferous aether** or **ether**" does not exist

Classification of waves based on the relationship between the displacement (oscillations) and direction of propagation: **longitudinal** (when both directions are parallel) or **transverse** (when both directions are perpendicular). Electromagnetic wave is a transverse one.





Perturbation can be described as:

$$y(x,t) = y_m \sin \underbrace{(kx - \omega t)}_{\text{phase}}$$

amplitude

phase

Angular frequency $\omega = \dfrac{2\pi}{T}$

(Angular) wave number - k

<u>Wavelength</u> λ – is a distance (measured along a direction of propagation) between the consecutive repetitions of the wave shape.

At t=0, the shape of wave is described as: $y(x,0) = y_m \sin(kx)$

From the definition of wavelength: $y(x_1, t) = y(x_1 + \lambda, t)$

hence: $y_m \sin(kx_1) = y_m \sin k(x_1 + \lambda)$

$$\Downarrow$$

$$k\lambda = 2\pi$$

Relationship between the wave number k and wavelength λ

$$k = \frac{2\pi}{\lambda}$$

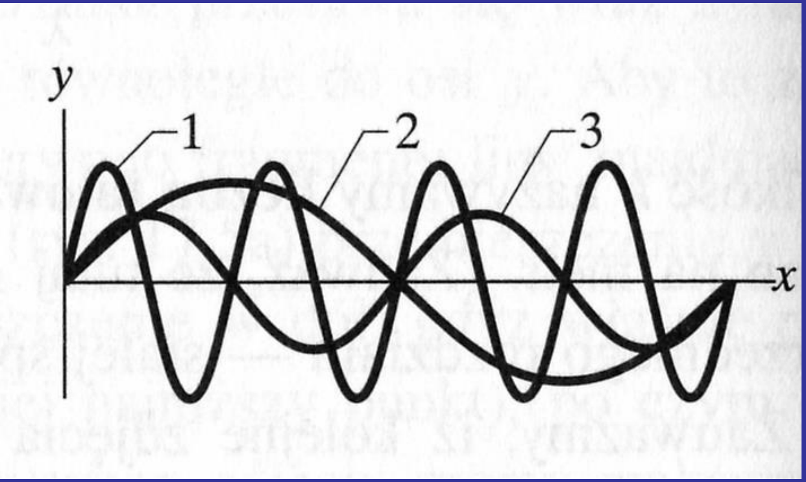
In 3D space:

$$y(\vec{\mathbf{r}}, t) = y_m \sin(\vec{\mathbf{k}} \circ \vec{\mathbf{r}} - \omega t)$$

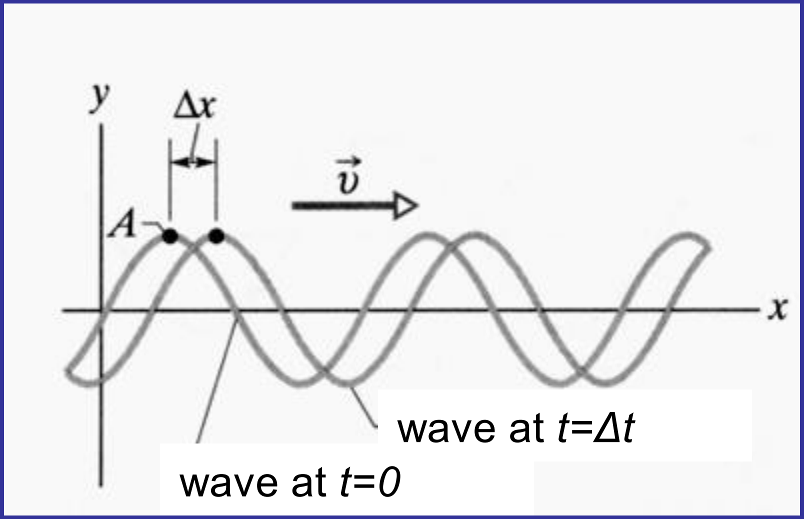
$\vec{\mathbf{k}}$ wave vector

Homework 12.1: Demonstrate that the above formula for $y(\vec{\mathbf{r}}, t)$ reduces to $y(x, t) = y_m \sin(kx - \omega t)$ in 1D space

Homework 12.2: The figure is a composite of three snapshots, each of a wave travelling along a particular string. The phases for the waves are given by: (a) 2x-4t, (b) 4x-8t, (c) 8x-16t. Which phase corresponds to which wave 1,2,3 in the figure ?



The speed of travelling wave



As the wave moves, each point of the moving wave, such as point A marked on a peak, retains its displacement y. Thus the phase in the displacement must remain a constant:

$$kx - \omega t = \mathrm{const}$$

As t increases, x increases as well to keep the argument constant.

This confirms that the wave pattern $y(x,t) = y_m \sin(kx - \omega t)$

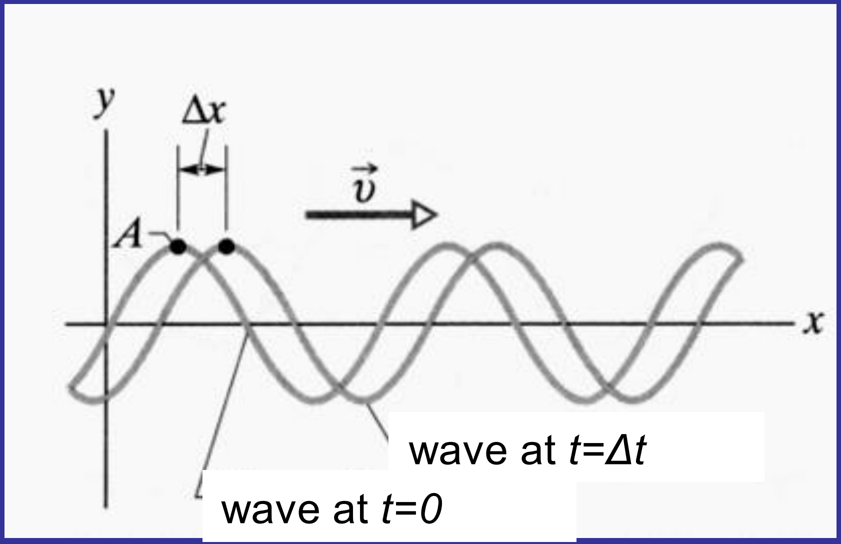
represents the wave propagating in positive x-axis direction

By analogy $y(x,t) = y_m \sin(kx + \omega t)$ represents the wave propagating in the opposite direction

CLASSIFICATION OF WAVES

According to shape of the wavefront, we can distinguish, e.g., spherical and plane waves. <u>Wavefront</u> is to a surface of points of the same phase of perturbation

Speed of the travelling wave



$$kx - \omega t = \text{const}$$

$$\frac{d}{dt}(kx - \omega t) = 0$$

$$k\frac{dx}{dt} - \omega = 0$$

$$kv - \omega = 0$$

Phase velocity (speed) v of the wave

$$v = \frac{dx}{dt}$$

$$v = \frac{\omega}{k}$$

$$v = \frac{\lambda}{T}$$

Speed of the wave depends on …

Speed of the mechanical wave is determined by inertia and elastic properties of the medium

Example 12.1 Speed of wave in a string.

Inertia: mass per unit length $\mu = M/L$ [kg/m]

Elastic properties: tensile strength T [kg m/s^2]

Analysis based on dimentionality gives as the only solution:

$$v = \sqrt{\frac{T}{\mu}}$$

Speed of mechanical waves in solids:

$$v = \sqrt{\frac{E}{\rho}}$$

Young's modulus

density

Speed of acoustic (sound) waves in gas:

$$v = \sqrt{\frac{B}{\rho}}$$

bulk modulus

density

$$B = -\frac{\Delta p}{\Delta V / V}$$

$$\boxed{v = \sqrt{\frac{\kappa p}{\rho}}}$$

pressure

$$\kappa = \frac{c_p}{c_v}$$

Speed of electromagnetic wave in vacuum:

$$c \approx 3 \cdot 10^{8}\,\mathrm{m/s}$$

Theory (Maxwell's equations) predicts

$$c = \frac{1}{\sqrt{\mu_o \varepsilon_o}}$$

universal constants

$$\mu_o = 1.26 \cdot 10^{-6}\, H/m$$

$$\varepsilon_o = 8.85 \cdot 10^{-12}\, F/m$$

In a medium (glass, water)

$$v = \frac{c}{n}$$

n – refractive index of a medium

GENERAL DIFFERENTIAL EQUATION OF A WAVE

Formula $y(x,t) = y_m \sin(kx - \omega t)$

is similar to the solution of the general equation of the harmonic oscillator

Looking for a general equation this formula solves?

$$\frac{\partial^2 y}{\partial t^2} = -\omega^2 y_m \sin(kx - \omega t) = -\omega^2 y$$

$$\omega = vk$$

$$\frac{\partial^2 y}{\partial x^2} = -k^2 y_m \sin(kx - \omega t) = -k^2 y$$

$$\frac{\partial^2 y}{\partial x^2} = \frac{1}{v^2}\frac{\partial^2 y}{\partial t^2}$$

GENERAL DIFFERENTIAL EQUATION OF A WAVE in 3D

Perturbation is described by a wave function Ψ(x,y,z,t)

$$\frac{\partial^2\Psi}{\partial x^2}+\frac{\partial^2\Psi}{\partial y^2}+\frac{\partial^2\Psi}{\partial z^2}=\frac{1}{v^2}\frac{\partial^2\Psi}{\partial t^2}$$

hence

$$\Delta\Psi(\vec{\mathbf{r}},t)=\frac{1}{v^2}\frac{\partial^2\Psi}{\partial t^2}$$

Differential Laplace operator (laplasian)

$$\Delta=\nabla\circ\nabla=\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}+\frac{\partial^2}{\partial z^2}$$

Any function in a form of $y = f(x \pm v t)$ is a solution od the more general wave equation:

$$\frac{\partial^2 y}{\partial x^2} = \frac{1}{v^2}\frac{\partial^2 y}{\partial t^2}$$

Minus sign „-" indicates the wave propagating in the positive x direction,

„+" sign corresponds to the negative direction of x

Homework 12.3. Propose other than $y(x,t) = y_m \sin(kx - \omega t)$ solutions of the general, differential wave equation

Energy density and intensity of a wave

Average energy density

$$< \rho_E >= b(\lambda) y_m{}^2$$

b(λ) is different for each type of a wave and depends on its wavelength

wave amplitude

Intensity of a wave

$$I = v < \rho_E >= b(\lambda) v\, y_m{}^2$$

energy flow in a unit of time per unit area, [I] =1 W/m^2

speed of a wave

Average power, average rate at which the energy is transported by a wave (for transverse wave on a string)

$$< P > = 2\left\langle \frac{dE_k}{dt} \right\rangle = \frac{1}{2}\mu v \omega^2 y_m^2$$

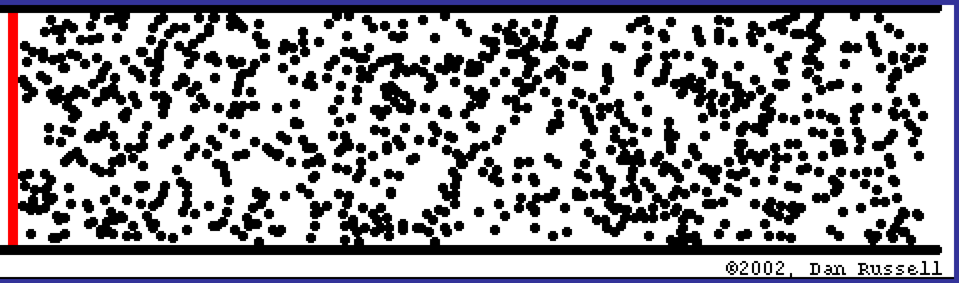
Parameters such as μ and v depend on a material and compressive strength of a string, in contrast to ω and y_m that are affected by the way in which the wave is generated

Dependence of the average power of the wave on the square of its amplitude and the square of its frequency has a general character and is valid for all types of waves.

HOMEWORK 12.4

A string has linear density μ=525 g/m and tension T=45N. We send a sinusoidal wave with frequency f=120 Hz and amplitude y_m=8.5 mm along the string. At what average rate does the wave transport energy?

Acoustic (sound) wave (longitudinal)



Displacement of the layer of fluid

$$s(x,t) = s_m \cos(kx - \omega t)$$

Pressure change in the air column

$$\Delta p(x,t) = \Delta p_m \sin(kx - \omega t)$$

$$\Delta p_m = (v\rho\omega)s_m$$

pressure amplitude

phase velocity

density of fluid

angular frequency

displacement amplitude

Example 12.2: The maximum pressure amplitude Δp_m that the human ear can tolerate in loud sounds is about 28 Pa (which is very much less than normal air pressure of 10^5 Pa). Find the amplitude of displacement s_m assuming the air density ρ=1.21 kg/m^3, at a frequency of 1000 Hz and sound speed 343 m/s

Given:

Δp_m= 28 Pa

ρ = 1.21 kg/m^3

f = 1000 Hz

v = 343 m/s

Solution:

$$s_m = \frac{\Delta p_m}{v\rho\omega} = \frac{\Delta p_m}{v\rho(2\pi f)}$$

To find:

s_m

Answer: s_m=11 μm

Conclusion:

The displacement amplitude of even the loudest sound that the ear can tolerate is very small.

HOMEWORK 12.5

Perform similar calculations to demonstrate that for the faintest detectable sound at 1000 Hz, the displacement amplitude is about 11 pm with the corresponding pressure of about $2.8 \cdot 10^{-5}$ Pa.

The ear is indeed a sensitive detector of sound waves

Intensity and sound level

The intensity I of a sound wave at a surface is the average rate per unit area at which the energy is transferred by the wave through or onto the surface.

$$I = \frac{P}{S}$$

time rate of energy transfer (power)

area of the surface intercepting the sound

For a wave emitted uniformly in all directions

$$I = \frac{P_s}{4\pi r^2}$$

power of the source

In analogy to the string

$$I = \frac{1}{2}\rho v \omega^2 s_m^2$$

The decibel scale

Human ear: displacement amplitude changes from 10^{-5} m for the loudest tolerated to 10^{-11} m for the faintest detectable sound; the ratio of these amplitudes is 10^6.

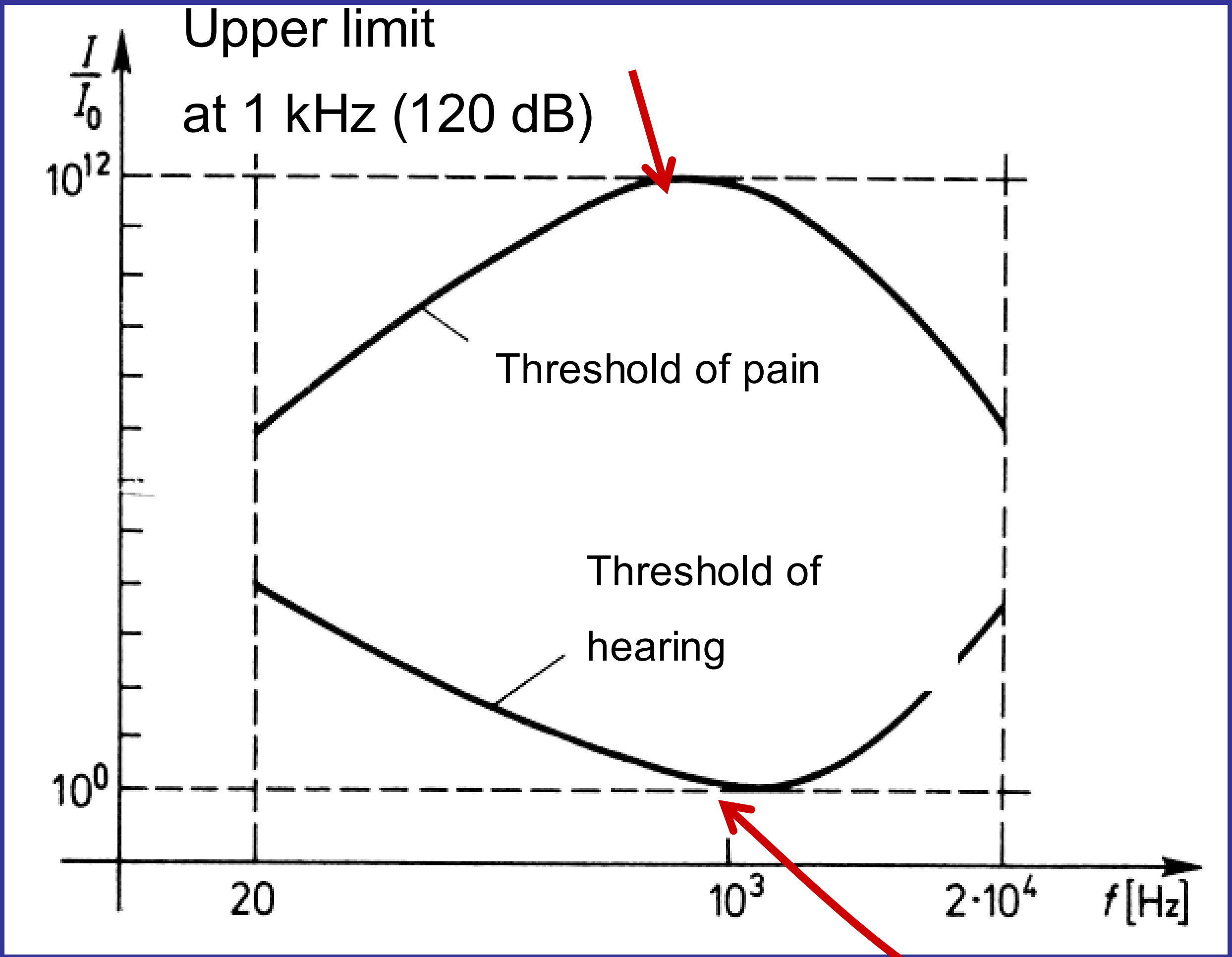
Intensity of sound is proportional to the square of amplitude, hence the range of intensities detected by human ear is quite wide - of about 10^{12} . Therefore, for comparing two intensities it is better to use logarithms.

$$y = \log x$$

$$y' = \log 10^{12} x = 12 + \log x = 12 + y$$

Sound level is determined by **Weber and Fechner** law.

Sensitivity curve for a human ear



Sound level

$$\Lambda = \eta \log \frac{I}{I_o}$$

η=1, unit is 1B (bel)
η=10, 1dB (decibel)

Intensity $I_o = 10^{-12}$ W/m^2 at 1 kHz is zero level (0 dB)

Human ear is characterized by different sensitivity at different frequencies of sound.

Decibel scale

$$\Lambda = (10\text{dB}) \log \frac{I}{I_o}$$

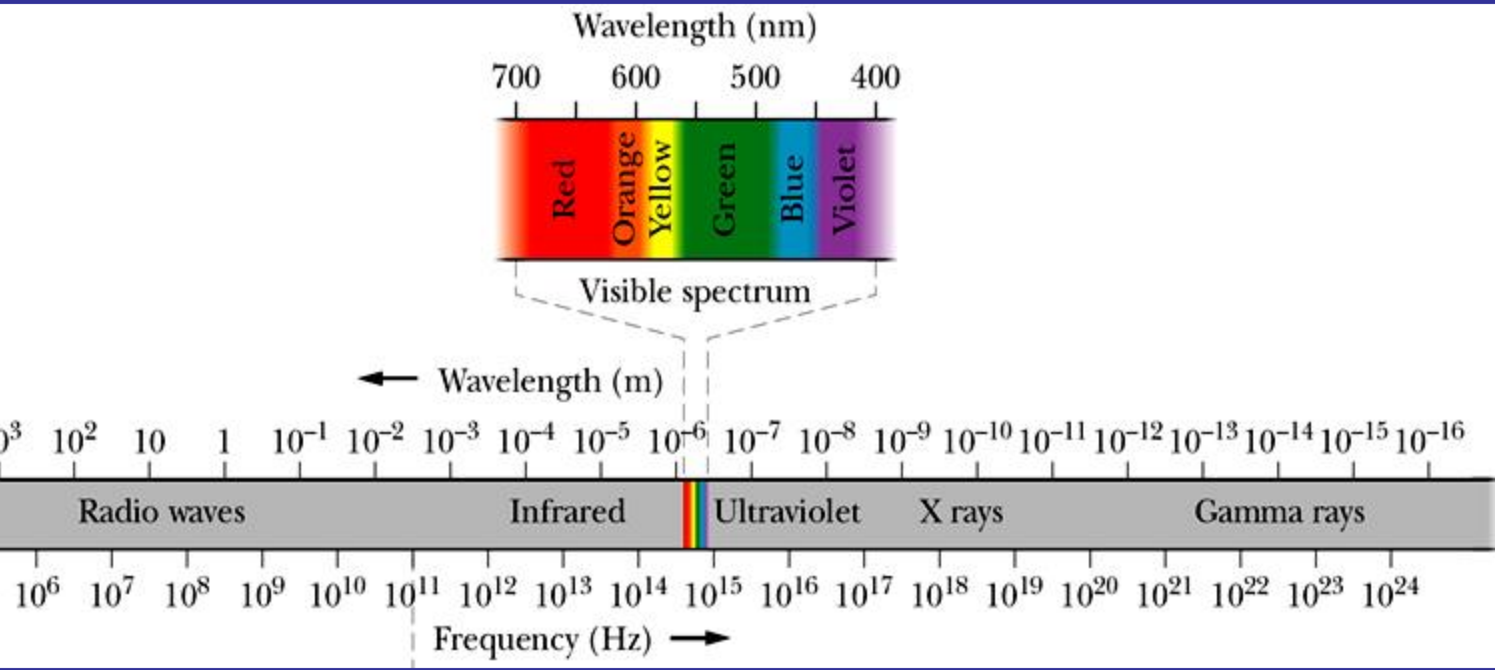
When the sound intensity I increases by one order of magnitude, i.e., 10 times, intensity level Λ increases by 10 dB

Hearing threshold	0 dB
Rustle of leaves	10 dB
Conversation	60 dB
Rock concert	110 dB
Pain threshold	120 dB
Jet engine	130 dB

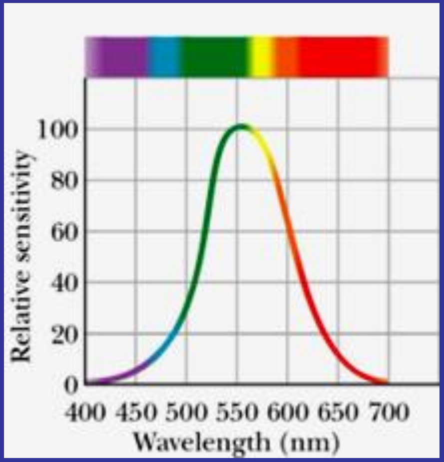
Electromagnetic radiation spectrum

Maxwell's rainbow

$$\nu = \frac{c}{\lambda}$$



Relative sensitivity of the average human eye



MAXWELL'S EQUATIONS

Law:	Integral form	Differential form
Gauss for electric field	$\oint_S \vec{\mathrm{E}} \circ d\vec{\mathbf{A}} = \dfrac{q}{\varepsilon_0}$	$\mathrm{div}\, \vec{\mathbf{E}} = \dfrac{\rho}{\varepsilon_o}$
Gauss for magnetic field	$\oint_S \vec{\mathbf{B}} \circ d\vec{\mathbf{A}} = 0$	$\mathrm{div}\, \vec{\mathbf{B}} = 0$
Ampere-Maxwell's	$\oint_C \vec{\mathbf{B}} \circ d\vec{\mathbf{l}} = \mu_o(i + \varepsilon_o \dfrac{d\Phi_E}{dt})$	$\mathrm{rot}\, \vec{\mathbf{B}} = \mu_o(\vec{\mathbf{j}} + \varepsilon_o \dfrac{\partial \vec{\mathbf{E}}}{\partial t})$
Faraday's	$\oint_C \vec{\mathbf{E}} \circ d\vec{\mathbf{l}} = -\dfrac{d\Phi_B}{dt}$	$\mathrm{rot}\, \vec{\mathbf{E}} = -\dfrac{\partial \vec{\mathbf{B}}}{\partial t}$

ELECTROMAGNETIC WAVE in vacuum

- It is assumed that $j=0$, $\rho=0$
- Maxwell equations:

$$\mathrm{div}\ \vec{\mathbf{E}} = 0 \quad \Longrightarrow \quad \nabla \circ \vec{\mathbf{E}} = 0$$

$$\mathrm{div}\ \vec{\mathbf{B}} = 0 \quad \Longrightarrow \quad \nabla \circ \vec{\mathbf{B}} = 0$$

$$\mathrm{rot}\ \vec{\mathbf{B}} = \mu_o\, \varepsilon_o \frac{\partial \vec{\mathbf{E}}}{\partial t} \quad \Longrightarrow \quad \nabla \times \vec{\mathbf{B}} = \mu_o \varepsilon_o \frac{\partial \vec{\mathbf{E}}}{\partial t}$$

$$\mathrm{rot}\ \vec{\mathbf{E}} = -\frac{\partial \vec{\mathbf{B}}}{\partial t} \quad \Longrightarrow \quad \nabla \times \vec{\mathbf{E}} = -\frac{\partial \vec{\mathbf{B}}}{\partial t}$$

Derivation of the EB wave

$$\nabla\times(\nabla\times\vec{\mathbf{E}}) = -\nabla\times\frac{\partial\vec{\mathbf{B}}}{\partial t} = -\frac{\partial}{\partial t}\nabla\times\vec{\mathbf{B}}$$

but $\nabla\times\vec{\mathbf{B}} = \mu_o\varepsilon_o\frac{\partial\vec{\mathbf{E}}}{\partial t}$

hence:

$$\nabla\times(\nabla\times\vec{\mathbf{E}}) = -\mu_o\varepsilon_o\frac{\partial^2\vec{\mathbf{E}}}{\partial t^2} \qquad (1)$$

Using the vectorial identity

$$\vec{\mathbf{a}}\times(\vec{\mathbf{b}}\times\vec{\mathbf{c}}) = (\vec{\mathbf{a}}\circ\vec{\mathbf{c}})\vec{\mathbf{b}} - (\vec{\mathbf{a}}\circ\vec{\mathbf{b}})\vec{\mathbf{c}}$$

$$\nabla\times(\nabla\times\vec{E}) = (\underbrace{\nabla\circ\vec{E}}_{0})\nabla - \nabla^2\vec{E} \qquad (2)$$

Combining (1) and (2) we get:

$$\nabla^2\vec{\mathbf{E}} = \mu_o\varepsilon_o\frac{\partial^2\vec{\mathbf{E}}}{\partial t^2}$$

General differential equation

$$\nabla^2\Psi(\vec{\mathbf{r}},t) = \frac{1}{v^2}\frac{\partial^2\Psi}{\partial t^2}$$

VELOCITY OF ELECTROMAGNETIC WAVE IN VACUUM

For a magnetic field $$\nabla^2\vec{\mathbf{B}} = \mu_o\varepsilon_o \frac{\partial^2\vec{\mathbf{B}}}{\partial t^2}$$

an electric field $$\nabla^2\vec{\mathbf{E}} = \mu_o\varepsilon_o \frac{\partial^2\vec{\mathbf{E}}}{\partial t^2}$$

Perturbation ψ is the vector of the electric field **E** or magnetic field **B** and the phase velocity v is determined by the universal constants, only:

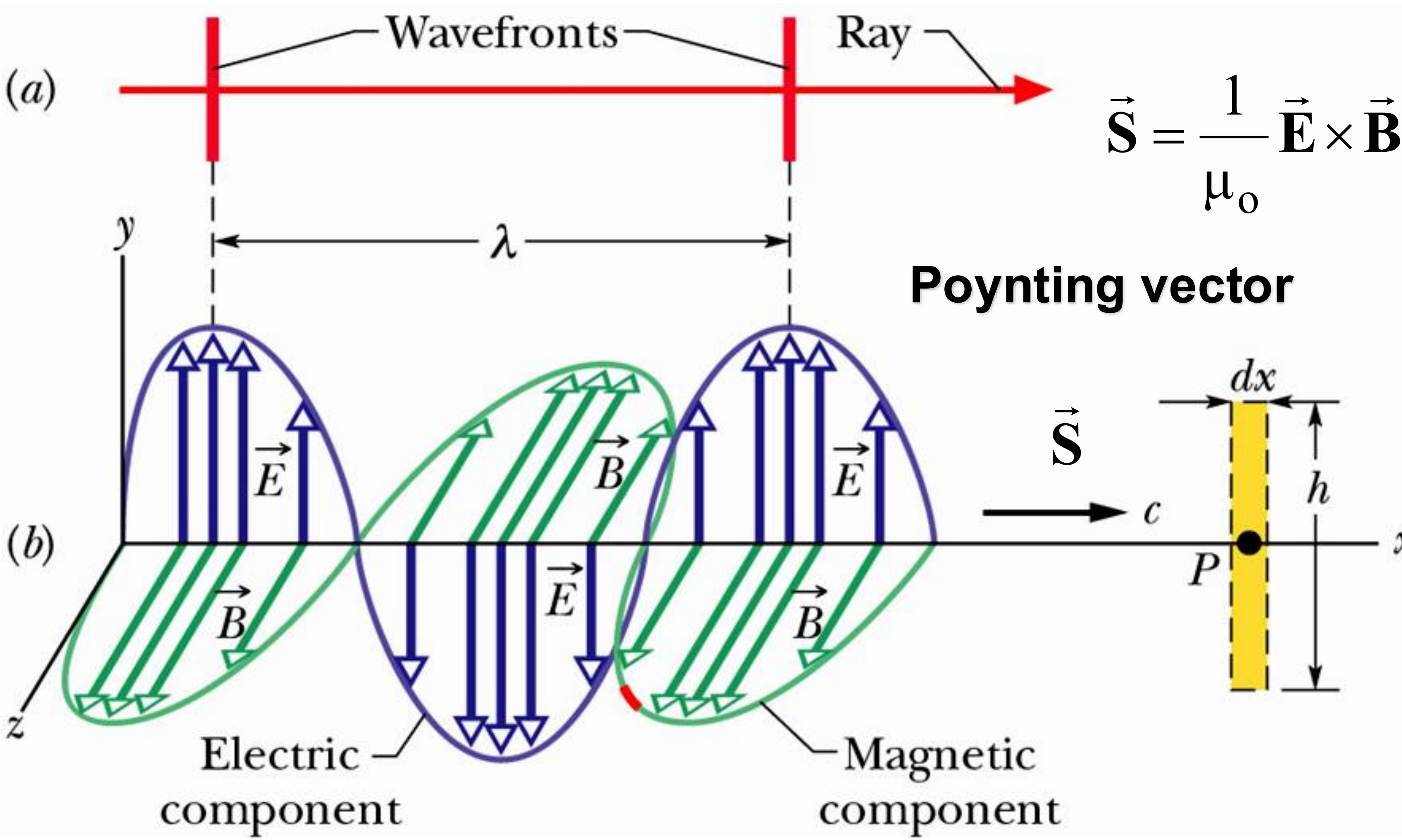
$$\mathrm{v} = \frac{1}{\sqrt{\mu_o\varepsilon_o}} = c$$

velocity of EB wave (light velocity) in vacuum can be calculated theoretically

Homework 12.6

- Prepare a topic: „Experimental methods of determination of light velocity in vacuum”

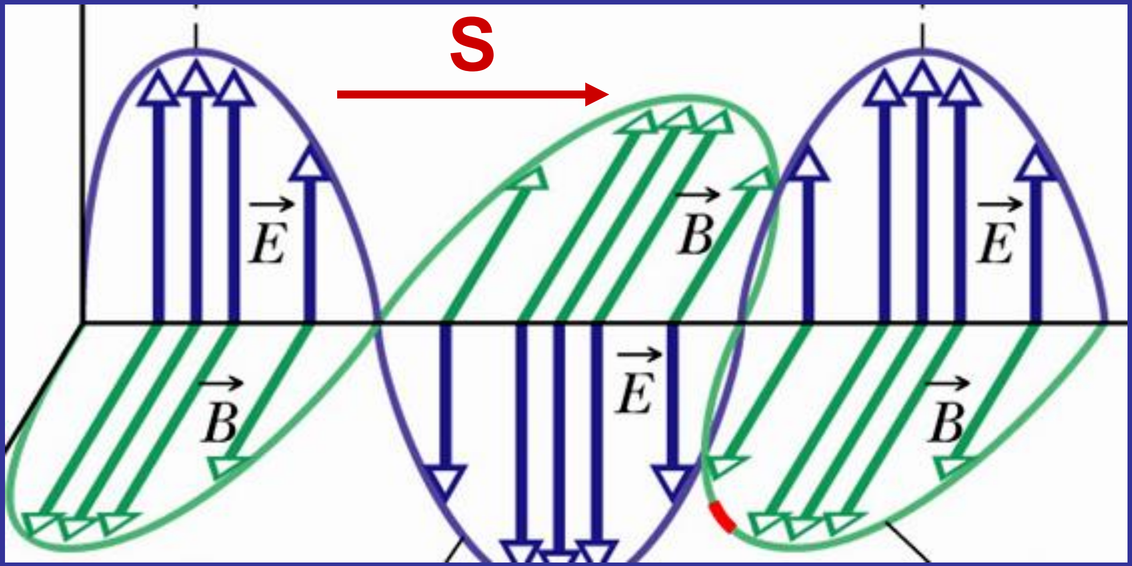
Propagation of electromagnetic wave



$$\vec{\mathbf{S}} = \frac{1}{\mu_o} \vec{\mathbf{E}} \times \vec{\mathbf{B}}$$

Poynting vector

Energy transfer – Poynting vector



Definition of Poynting vector

$$\vec{\mathbf{S}} = \frac{1}{\mu_o}\vec{\mathbf{E}} \times \vec{\mathbf{B}}$$

The direction of the Poynting vector of an electromagnetic wave at any point gives the wave's direction of travel and the direction of energy transport at that point.

Intensity of electromagnetic wave

Magnitude of the Poynting vector is related to the rate at which energy is transported per unit area. Time-averaged magnitude of the Poynting vector represents intensity of the electromagnetic wave.

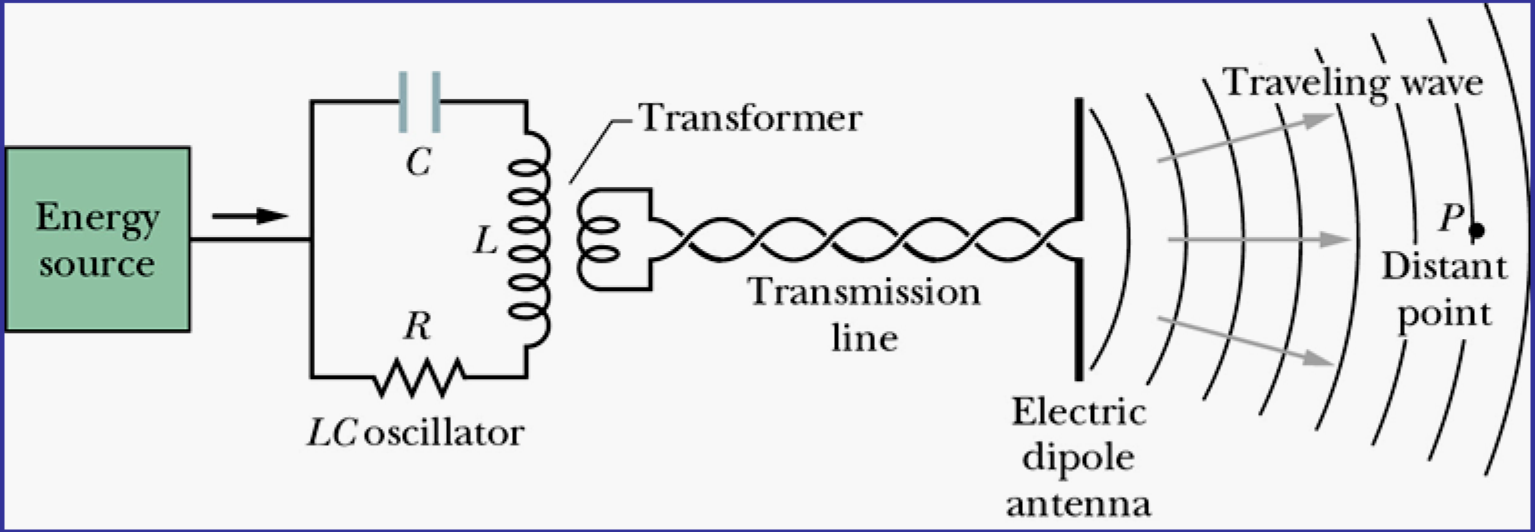
Instantaneous energy flow rate

$$S = \frac{1}{\mu_o} EB = \frac{1}{c\mu_o} E^2$$

Intensity of electromagnetic wave

$$I = S_{average} = \frac{1}{2c\mu_o} E_m^{\ 2}$$

Generation of electromagnetic wave in the shortwave radio region of the spectrum



$$E(x,t) = E_m \sin(kx - \omega t)$$

$$B(x,t) = B_m \sin(kx - \omega t)$$

$$\frac{E_m}{B_m} = c$$

$$\frac{E}{B} = c$$

H. Hertz (1888) experimental confirmation of existence of electromagnetic wave

FUNDAMENTAL WAVE PHENOMENA:

- **interference**
- **diffraction**
- **polarization**

but also: refraction, dispersion, reflection, transmission, absorption

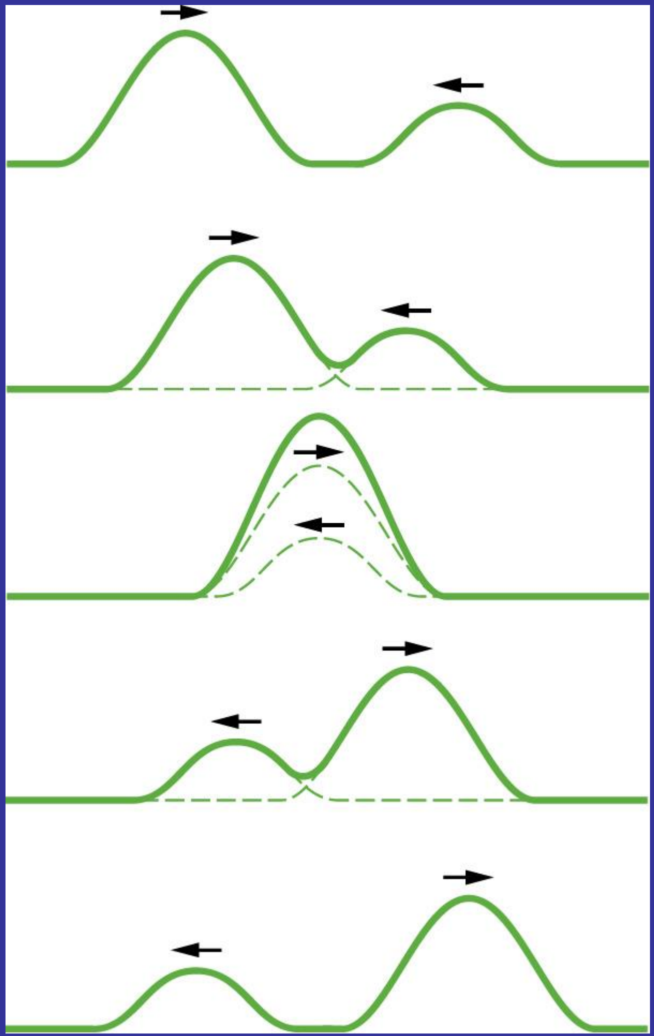
All these phenomena are common for all types of waves.

SUPERPOSITION OF WAVES

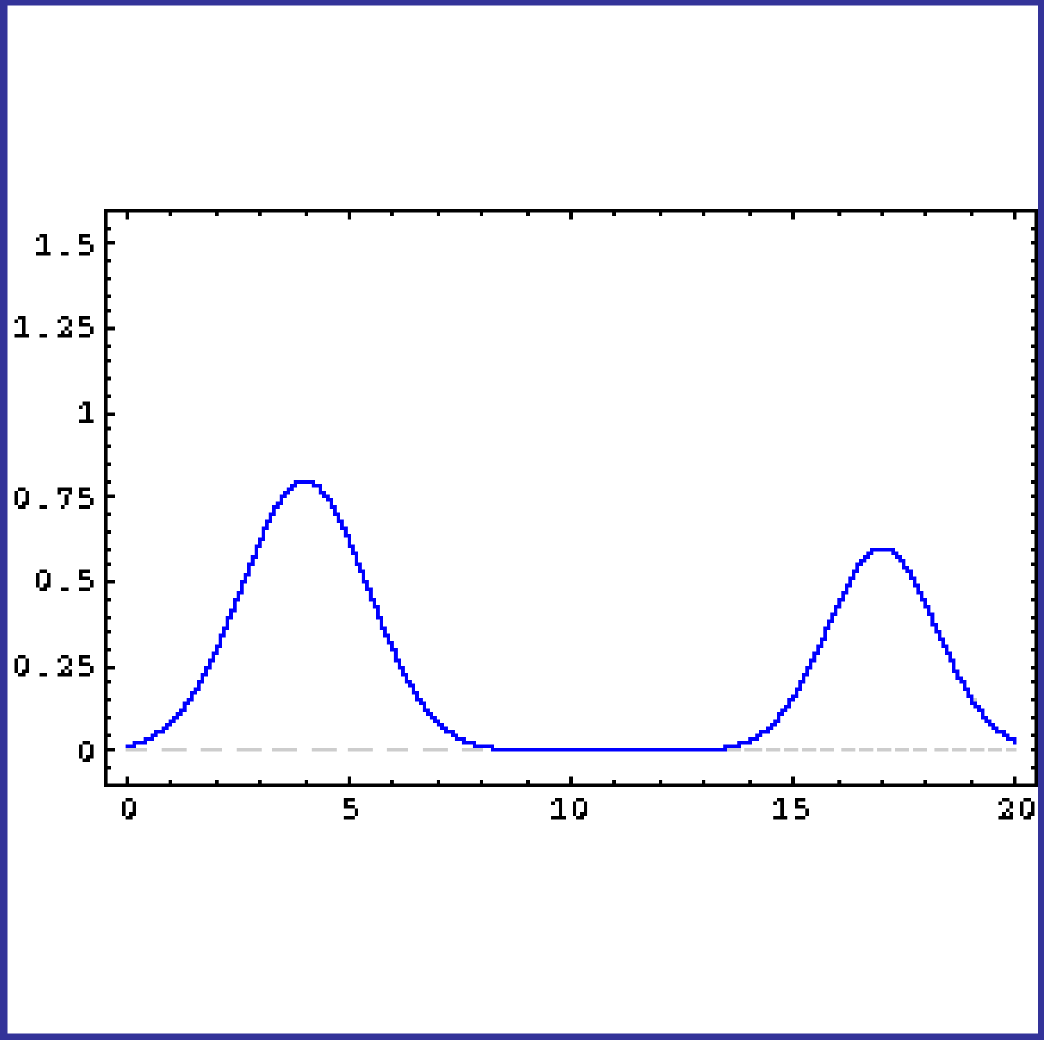
It often happens that two or more waves pass simultaneously through the same region. Overlapping waves add algebraically to produce a resultant wave - **net wave**.

$$y_w(x,t)=y_1(x,t)+y_2(x,t)$$

Overlapping waves do not in any way alter the travel of each other.

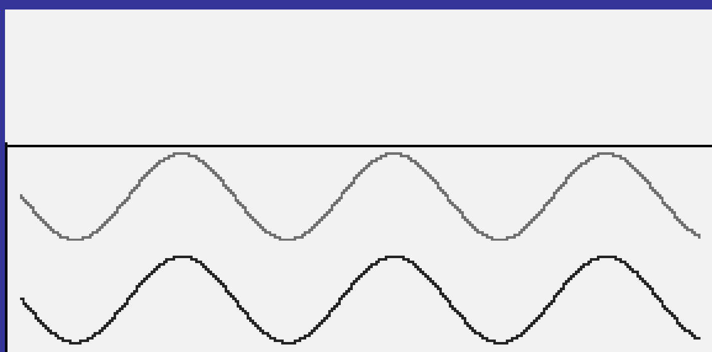


Demonstration of principle of superposition

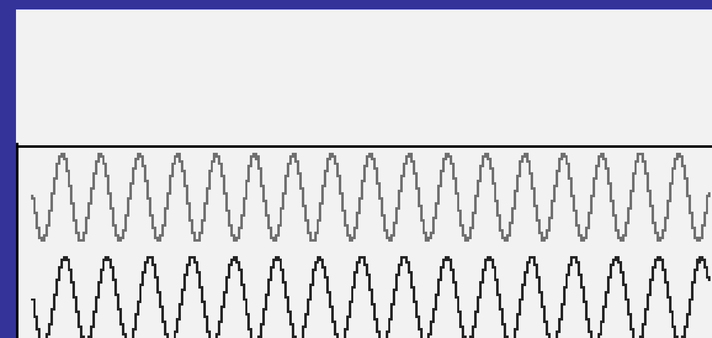


Consequences of superposition of waves

Enhancement (constructive interference) or attenuation (destructive interference)



Beats (superposition of waves with similar frequencies)



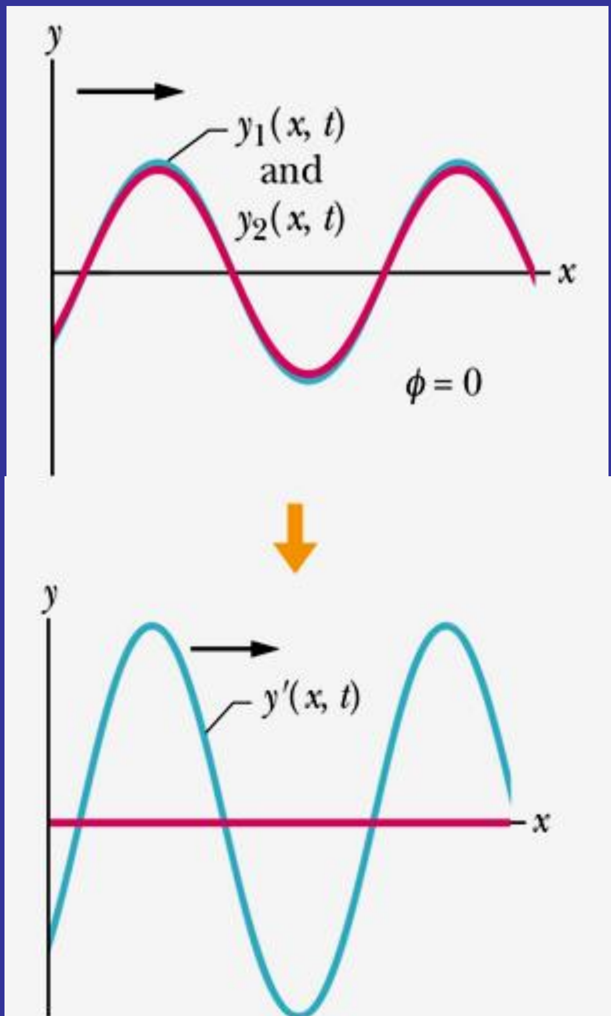
Interference

Assume that two sinusoidal waves of the same wavelength and amplitude travel along a stretched string in the same direction. Waves interfere to give the net sinusoidal wave travelling in the same direction. Amplitude of the net wave depends on the relative phase difference of interfering waves.

$$y_1(x,t) = y_m \sin(kx - \omega t)$$

$$y_2(x,t) = y_m \sin(kx - \omega t + \varphi)$$

$$y = y_1(x,t) + y_2(x,t) = \underbrace{\left[2 y_m \cos\frac{1}{2}\varphi\right]}_{\text{amplitude}} \sin(kx - \omega t + \frac{1}{2}\varphi)$$

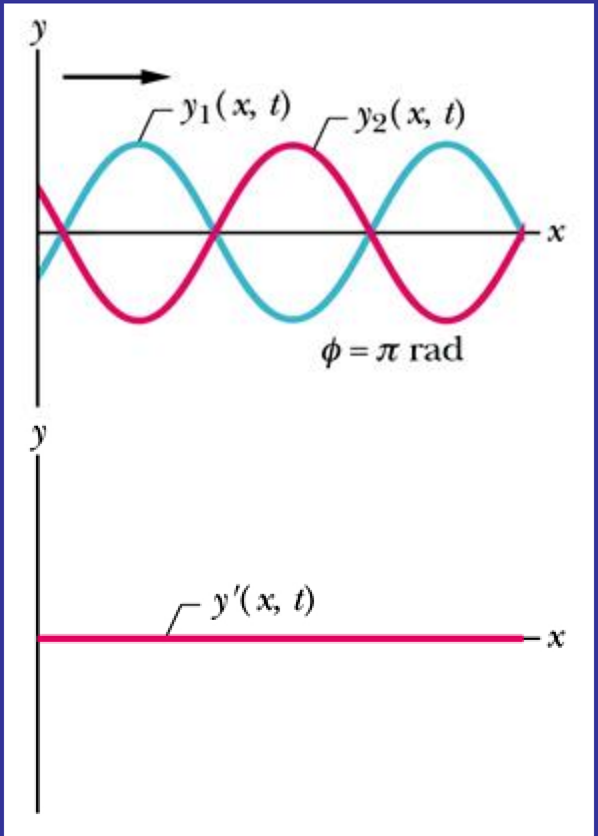


Constructive interference (enhancement) occurs when the waves are in phase, i.e., for φ=0, 2π, 4π, …

Amplitude of the net wave is twice the amplitude of the interfering waves.

$$y'_m = 2y_m \cos\frac{1}{2}\varphi = 2y_m$$

Intensity of the net wave is four times bigger than the amplitude of the interfering waves.



Destructive interference – complete attenuation occurs when in the waves are in the opposite phases (exactly out of phase), i.e., φ= π, 3π, 5π, …

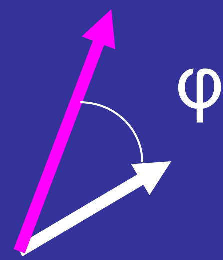
Amplitude and intensity of the resultant wave are zero.

$$y'_m = 2y_m \cos\frac{1}{2}\varphi = 0$$

<u>Reminder</u>: Similar effect was observed for oscillations in the same direction.

Phasor

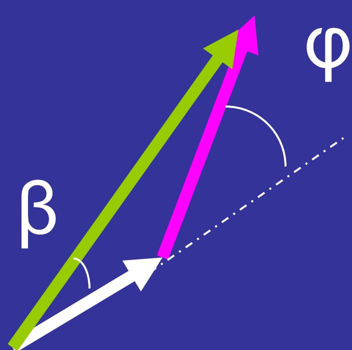
Phasor is a vector of length equal to the amplitude of the wave. Phasor rotates around an origin with an angular speed equal to the angular frequency of the wave, ω.



$$y_1(x,t) = y_{m1}\sin(kx - \omega t)$$

$$y_2(x,t) = y_{m2}\sin(kx - \omega t + \varphi)$$

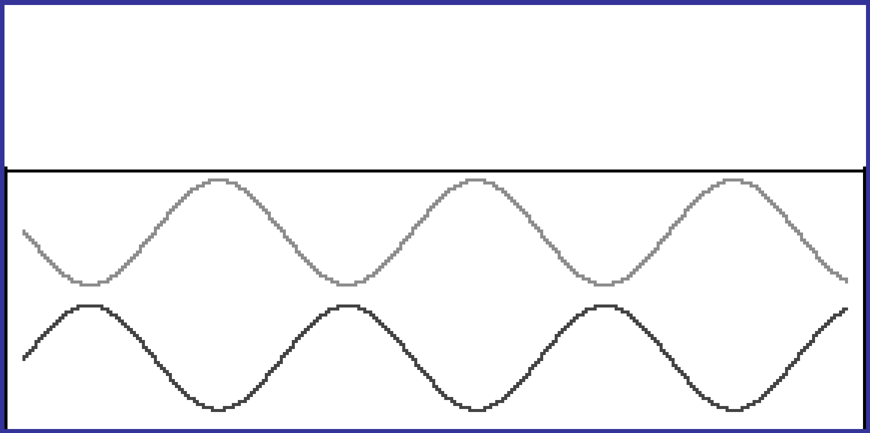
Result of interference– a net vector



$$y'(x,t) = y'_m \sin(kx - \omega t + \beta)$$

We can use phasors to combine waves *even if their amplitudes are different*

Standing wave



Standing wave is formed when two sinusoidal waves of the same wavelength and amplitude propagate along a stretched string in opposite directions

$$y_1(x,t) = y_m \sin(kx - \omega t)$$

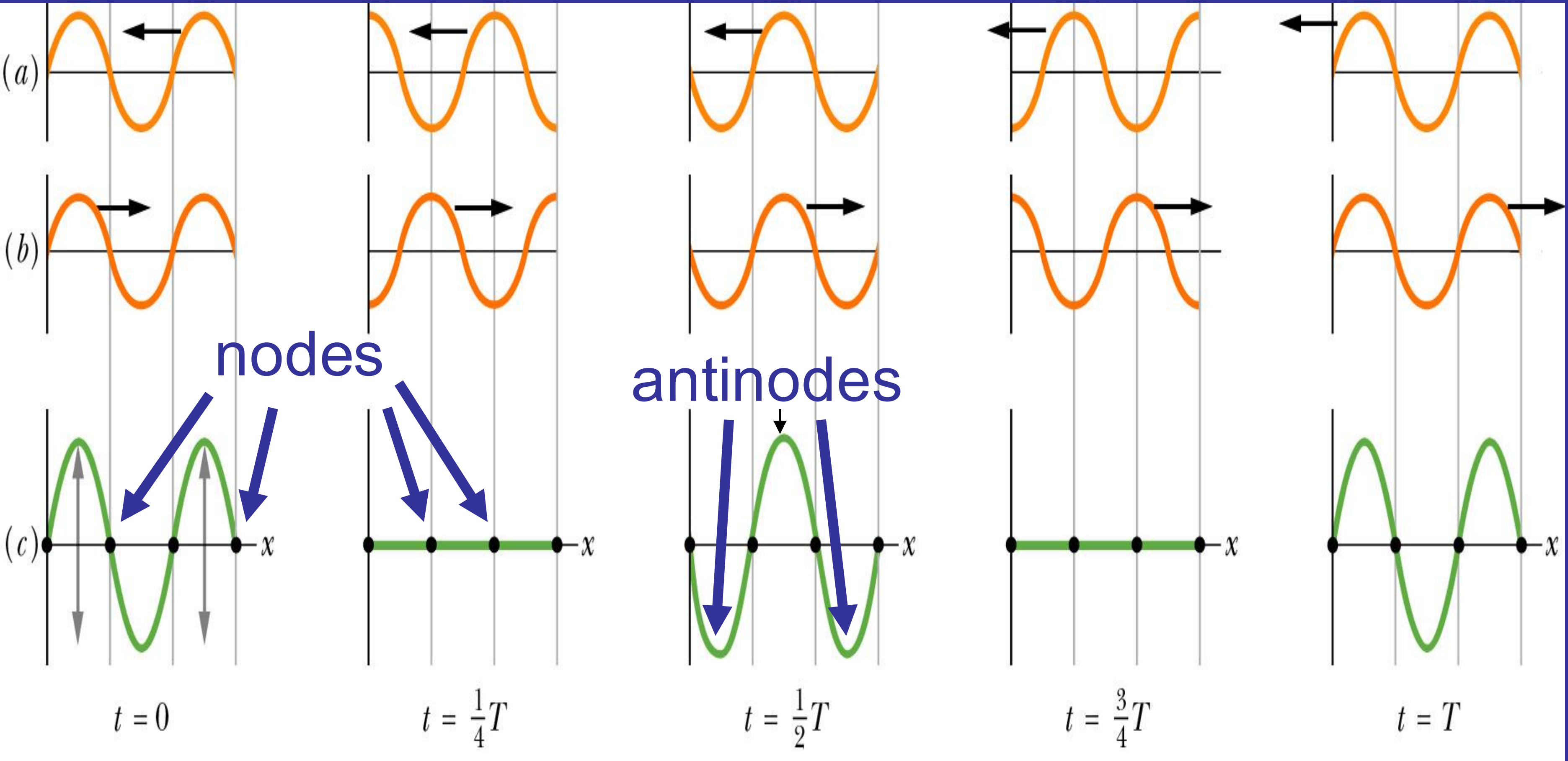
$$y_2(x,t) = y_m \sin(kx + \omega t)$$

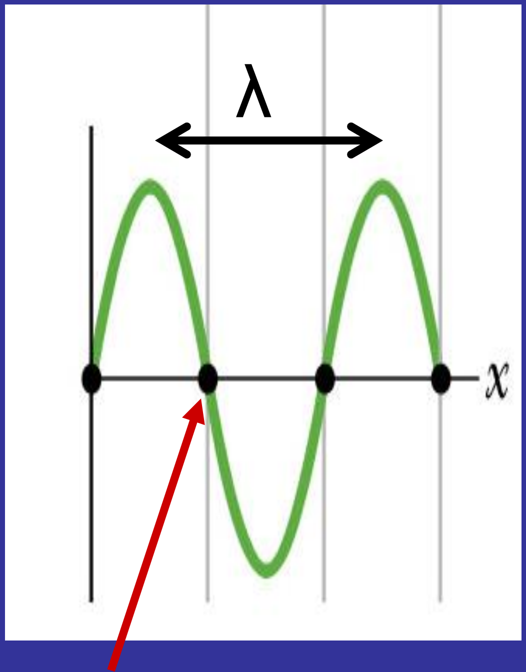
It can be shown that

$$y = y_1 + y_2 = \underbrace{\left[2y_m \sin kx\right]}_{\text{wave amplitude}}\cos(\omega t)$$

Standing wave

Positions of nodes and antinodes are fixed. Wave amplitude depends on position.





Node position for n'=1

Position of nodes is given by:

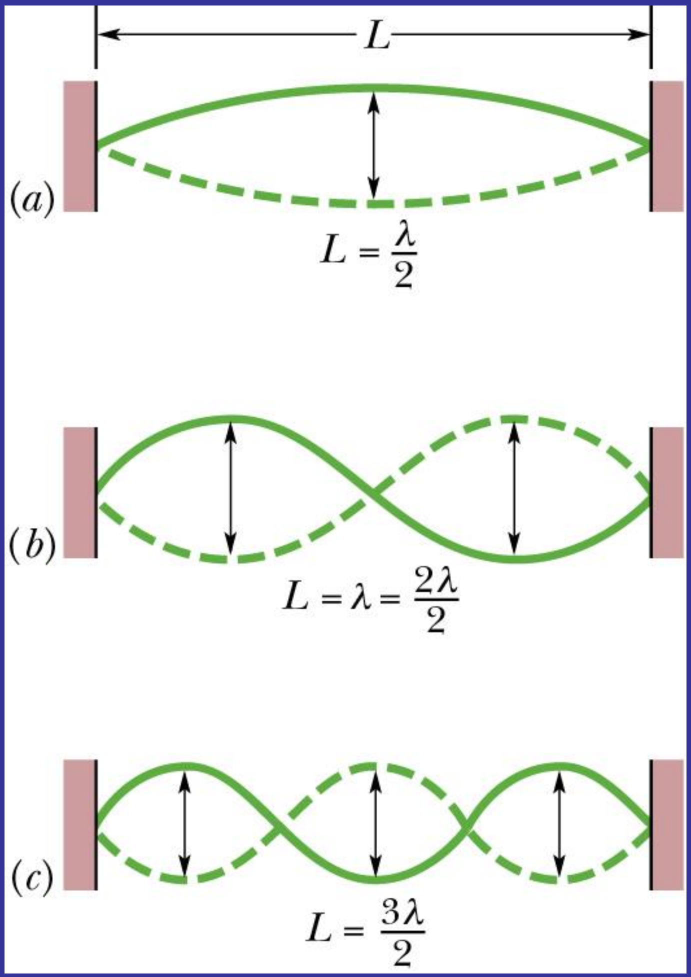
$$x = n'\frac{\lambda}{2}$$

where n'=0,1,2,….

Resonance occurs, when at certain frequencies a standing wave of a big amplitude is generated as a result of interference.

Each possible frequency, and the corresponding standing wave pattern is an oscillation mode.

Resonance



Imposing boundary conditions we introduce quantization of wavelength and frequency

Boundary conditions :

at x=0, y=0 and at x=L, y=0 (nodes at the fixed end points)

Quantization of wavelength:

$$\lambda_{n'} = \frac{2L}{n'}$$

where n'=1,2,3….

Quantization of frequency:

$$\gamma_{n'} = n' \frac{v}{2L}$$

wave speed

Resonant frequencies are integer multiples of the lowest frequency – fundamental mode of oscillations or the first harmonics γ_1

$$\gamma_1 = \frac{v}{2L}$$

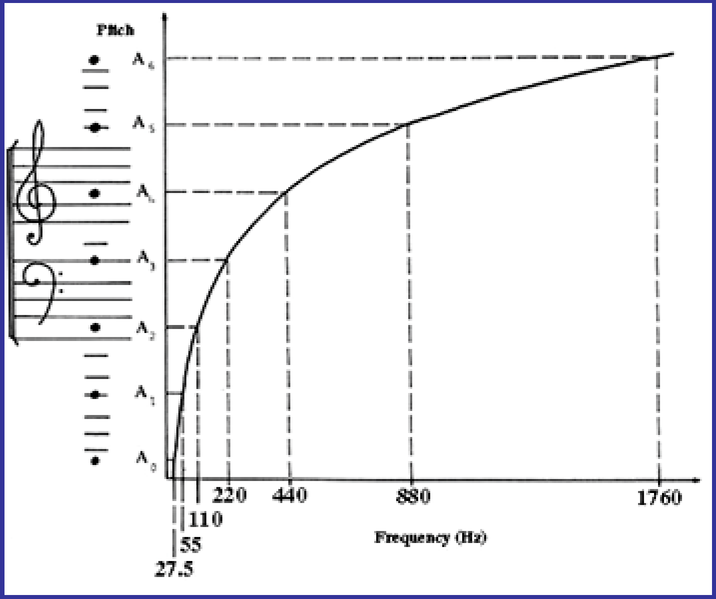
The collection of all possible oscillation modes is called the harmonic series and is described as:

$$\gamma_{n'} = n'\gamma_1$$

Harmonic number

Properties of sound

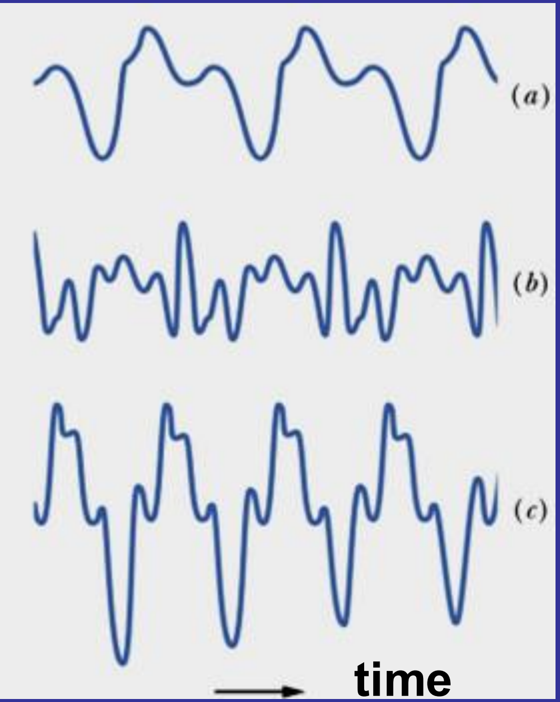
- pitch – frequency of the fundamental mode
- loudness – subjective measure, related to the amplitude and frequency
- timbre or quality – harmonic content



a) flute

b) oboe

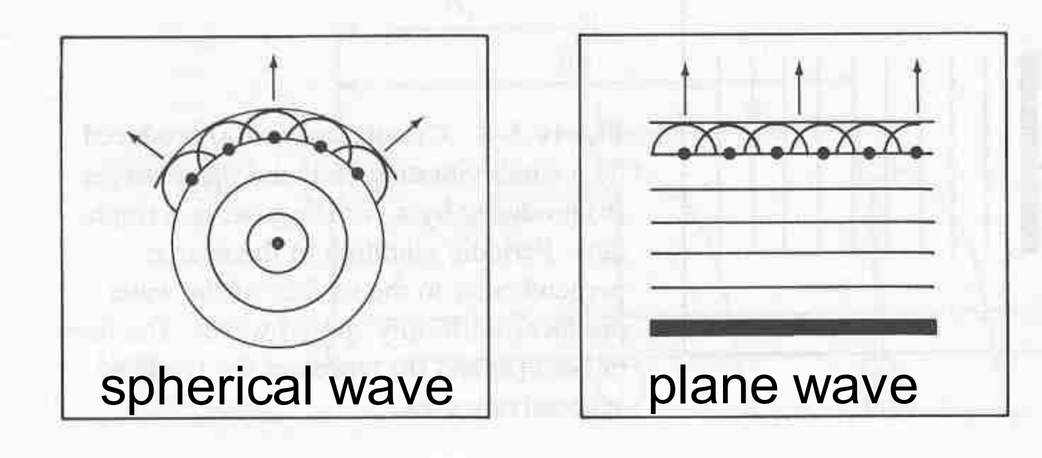
c) saxophone



Light – as a wave

Christian Huygens, Dutch scientist in 1678 advanced the first wave theory of light

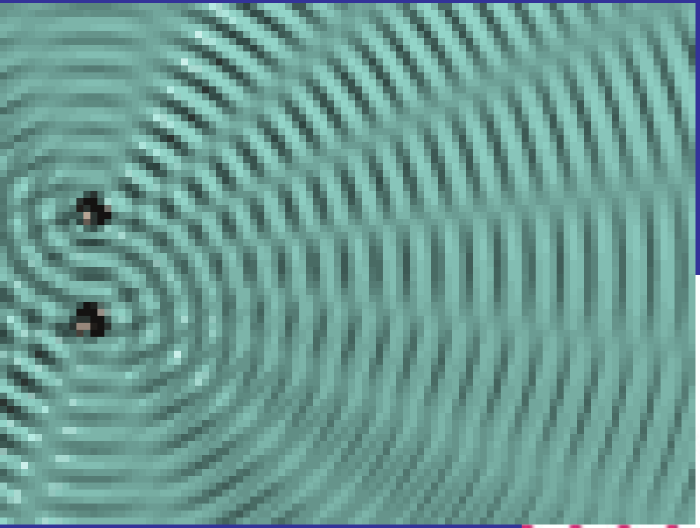
Huygens' principle: All points on a wavefront serve as point sources of spherical secondary wavelets. After a time t, the new position of the wavefront will be that of a surface tangent to these secondary wavelets.

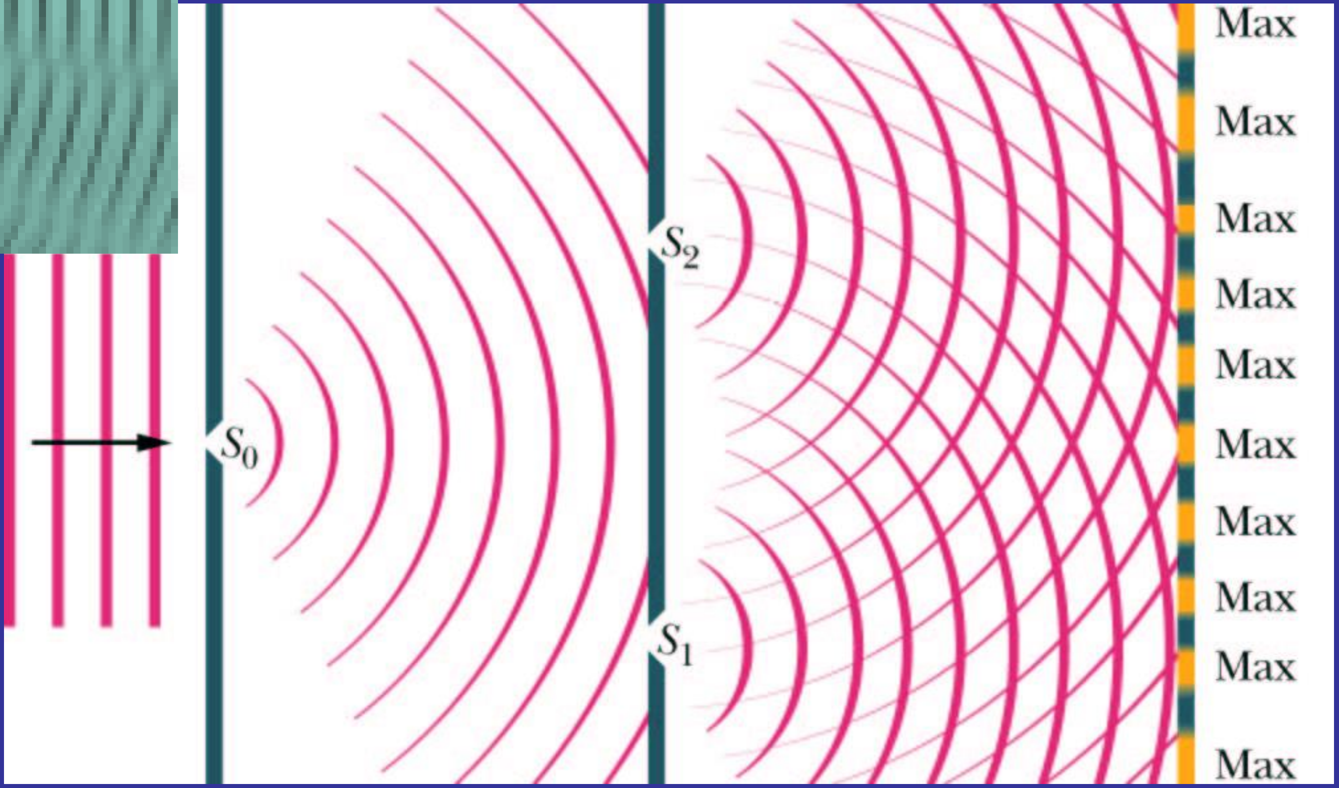


It allows to derive, e.g. the law of refraction, the law of reflection. It is useful in the interpretation of interference and diffraction.

Young's experiment

1801 – light is a wave because it undergoes interference





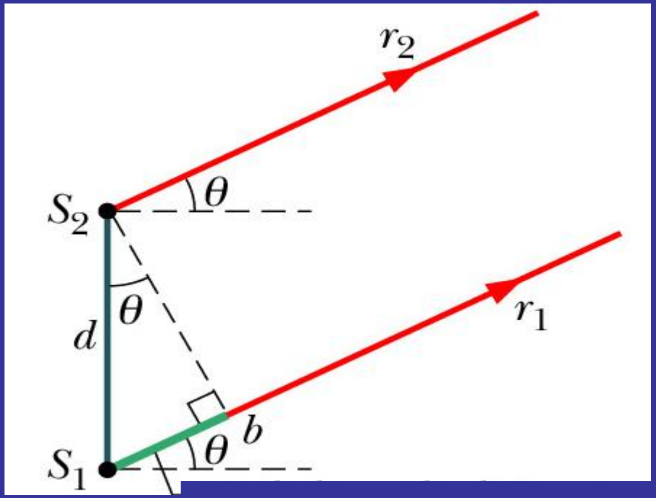
The result of the interference depends on the phase difference $\Delta\varphi$

The phase difference $\Delta\varphi$ between two waves change if the waves travel paths of different lengths ΔL.

$\Delta\varphi$ - 2π

ΔL - λ

$$\Delta\varphi = \frac{2\pi}{\lambda}\Delta L$$

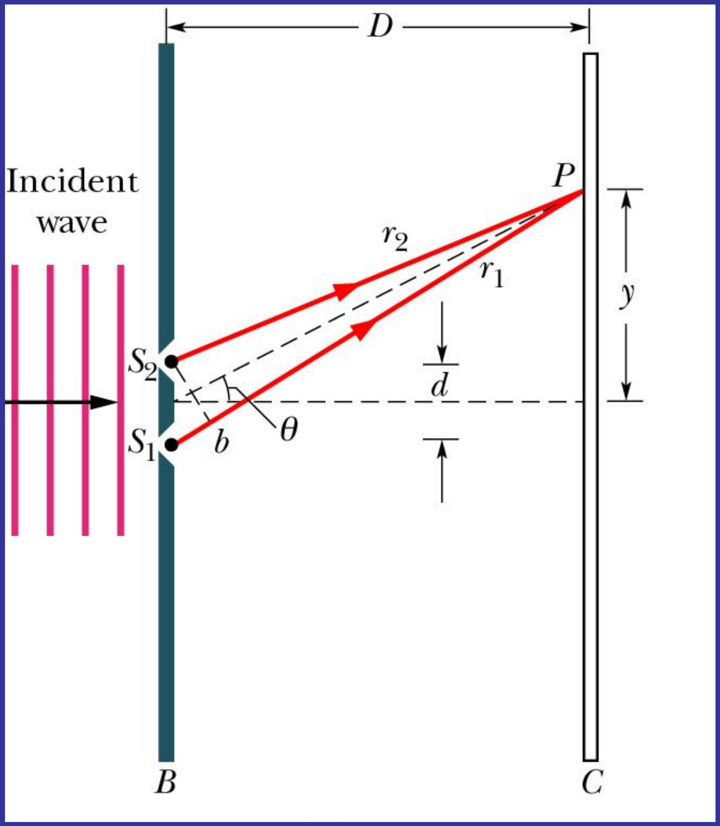


$S_1 b = \Delta L$

$$\Delta L = S_1 b = d\sin\theta$$

when $\Delta L=\lambda$ then $\Delta\varphi=2\pi$ and the constructing interference occurs

path length difference



Conditions for interference:

Phase difference has to be constant in time and space – coherence

Light sources have to be coherent in order to observe the interference

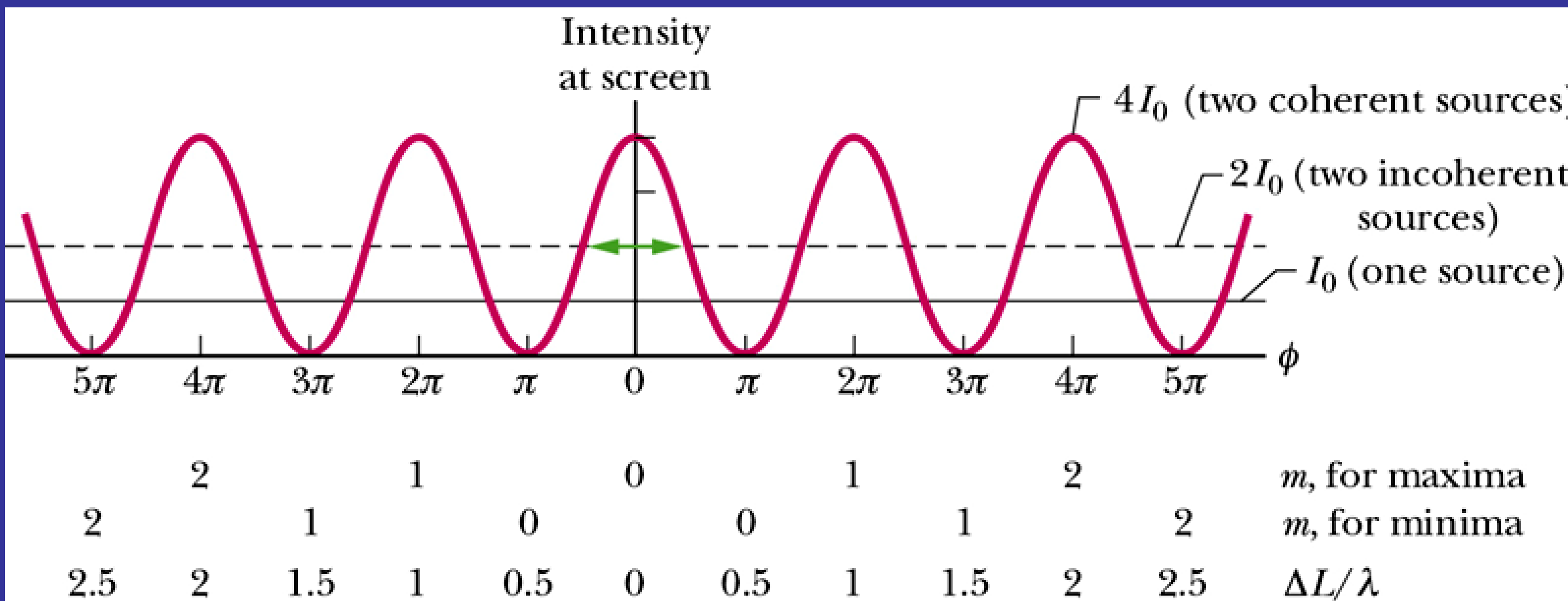
Constructive interference (maximum)

$$\boxed{d\sin\theta = m\lambda}$$ m=0,1,2,..

Destructive interference (minimum)

$$\boxed{d\sin\theta = (m+\frac{1}{2})\lambda}$$

Interference pattern– distribution of intensity at the far screen



$$I = 4I_o \cos^2(\varphi/2)$$

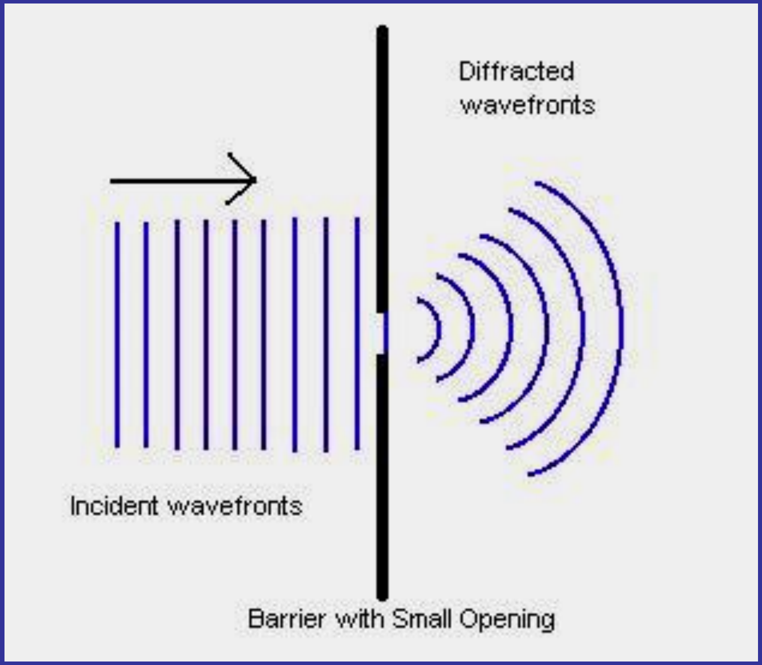
slit separation

phase difference

$$\varphi = \frac{2\pi d}{\lambda}\sin\theta$$

angle of observation

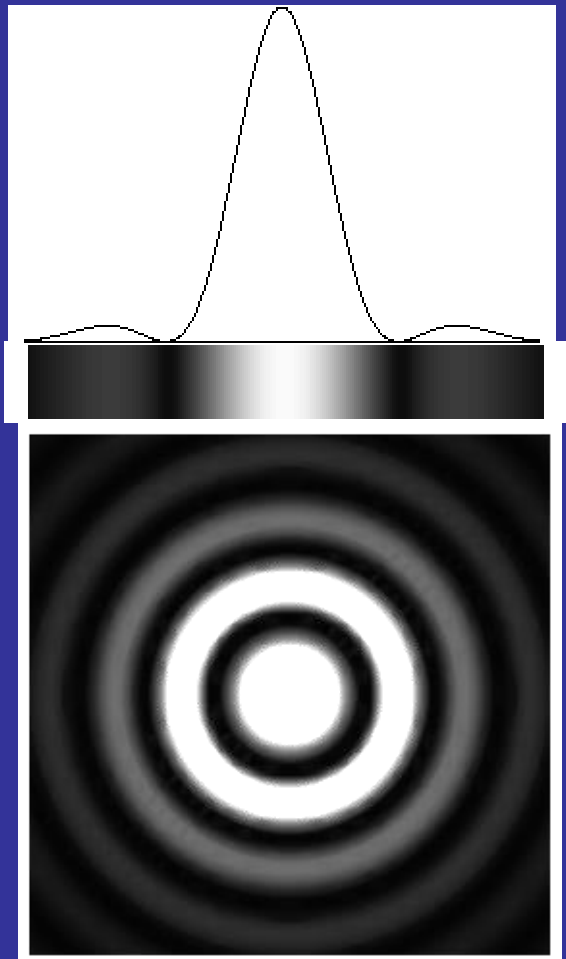
Diffraction

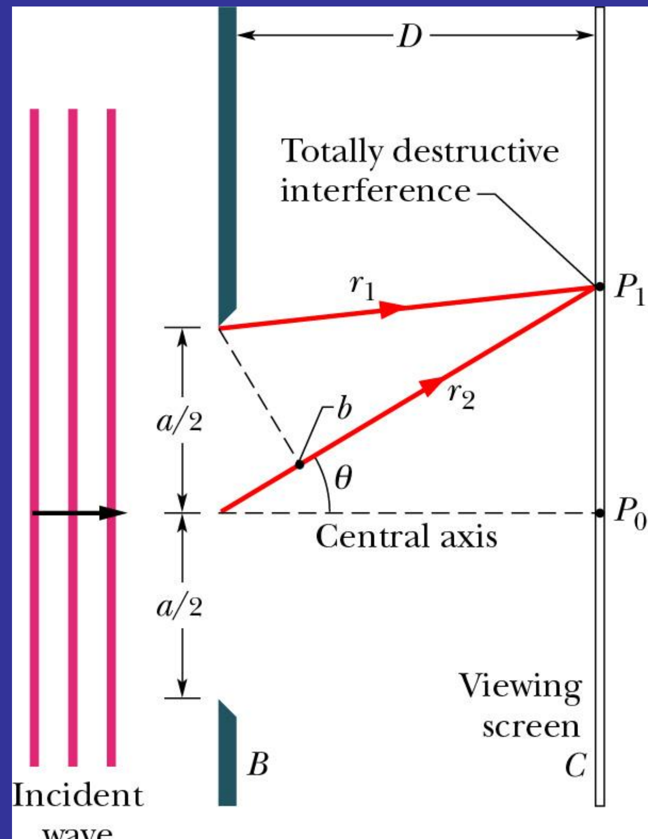


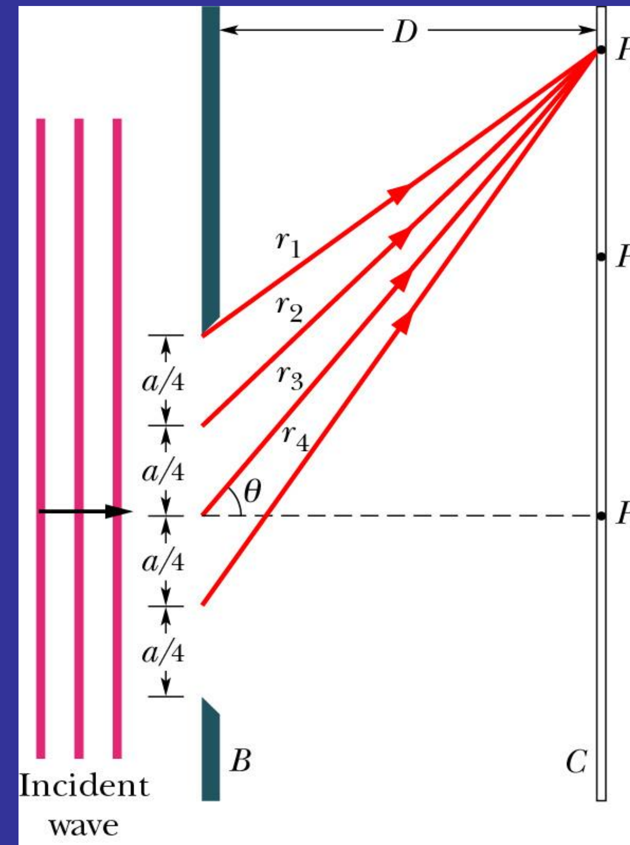
If a wave encounters a barrier that has an opening of dimensions similar to the wavelength, the part of the wave that passes through the opening will flare (spread) out – will diffract – into the region beyond the barrier.

As a result of the diffraction a regular pattern of bright and dark interference fringes appears on the screen – it is called a diffraction pattern.

Diffraction pattern of a single slit



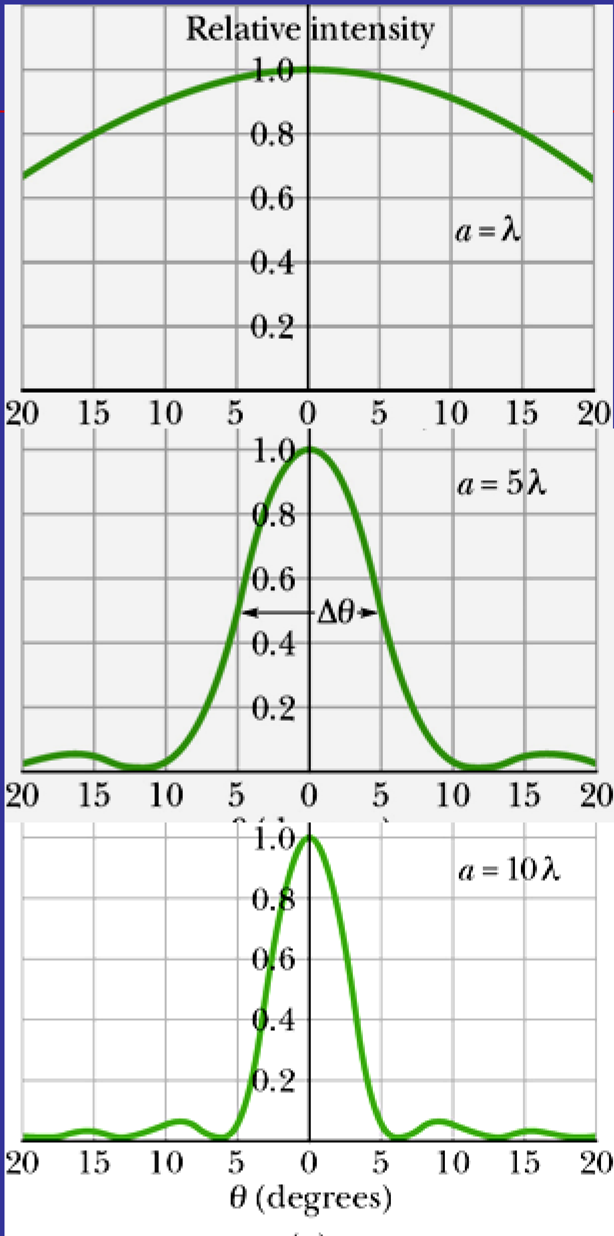




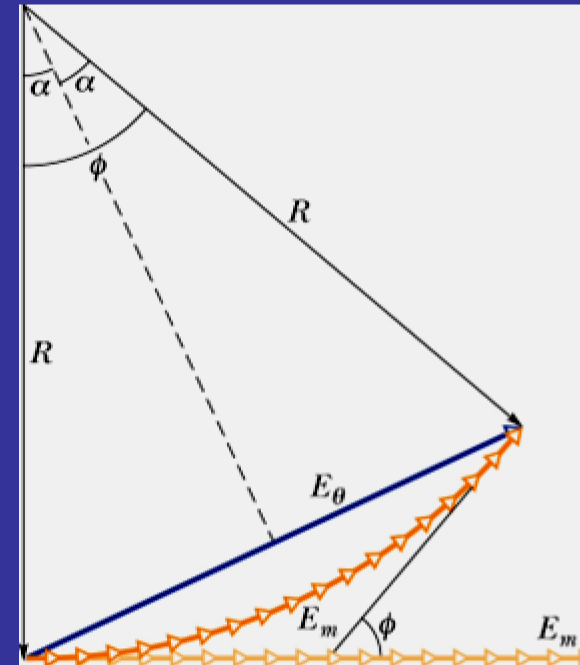
Dark fringes – minima appear when

Slit width $\quad \boxed{a \sin\theta = m\lambda} \quad$ m=0,1,2,..

Diffraction angle



Method of phasors – calculation of intensity in the case of diffraction at a single slit

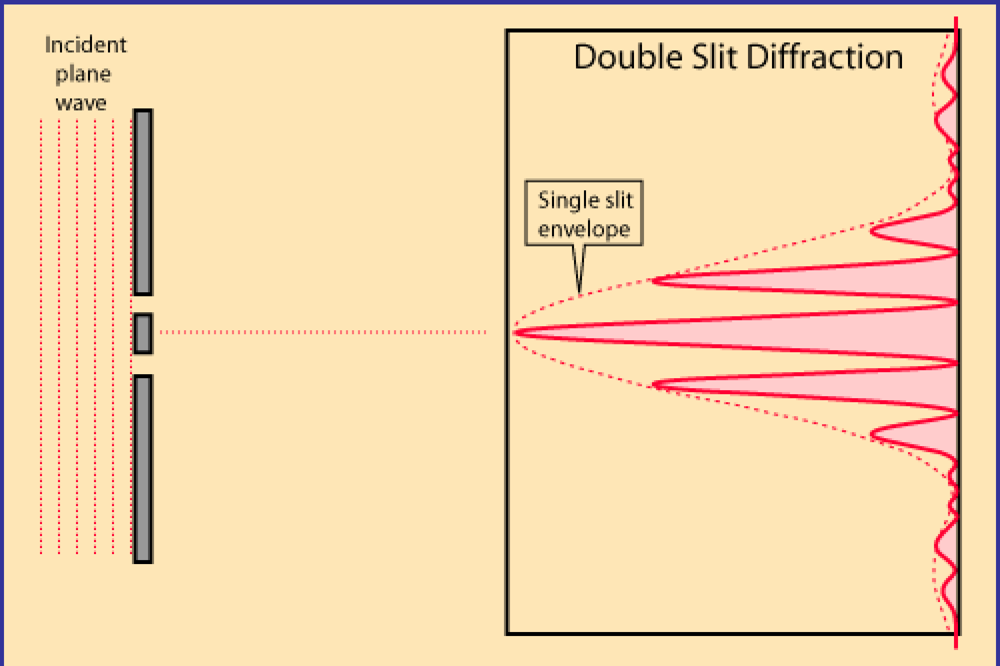


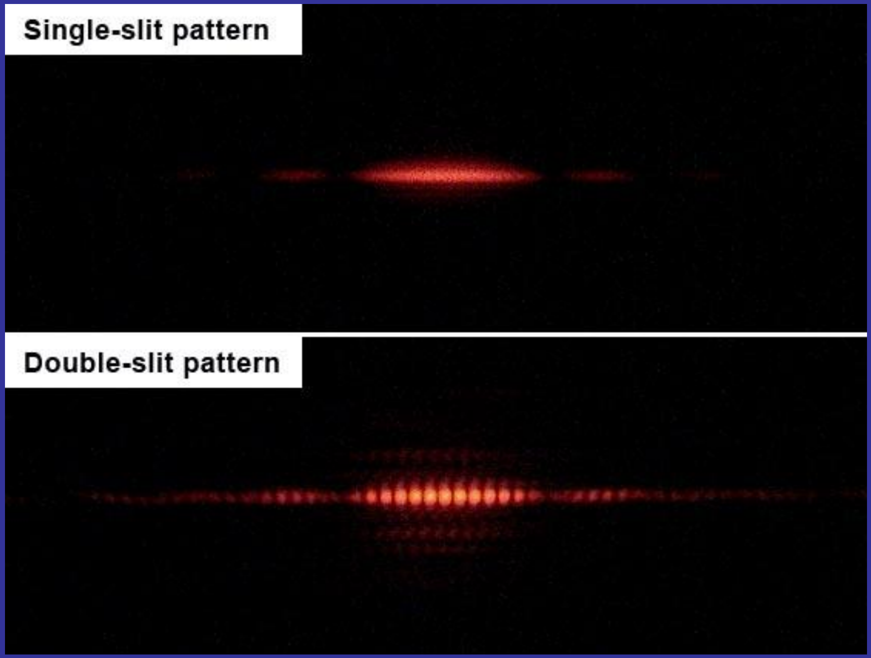
$$I(\theta) = I_o \left(\frac{\sin \alpha}{\alpha} \right)^2$$

$$\alpha = \frac{\varphi}{2} = \frac{\pi a}{\lambda} \sin \theta$$

When a/λ increases, the width of the central maximum decreases.

Interference pattern from two slits – diffraction at a single slit included



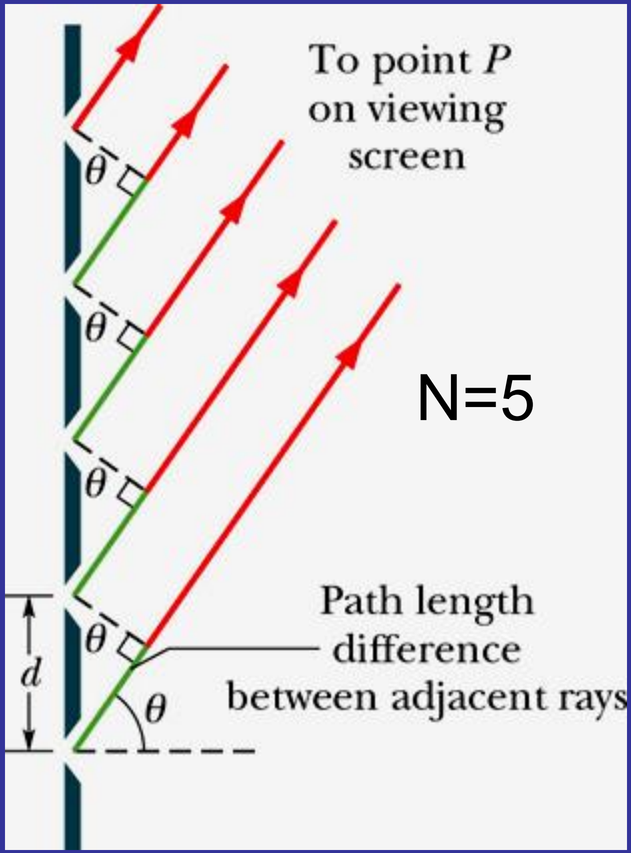


$$I(\theta) = 4I_o\left(\frac{\sin\alpha}{\alpha}\right)^2\cos^2\beta$$

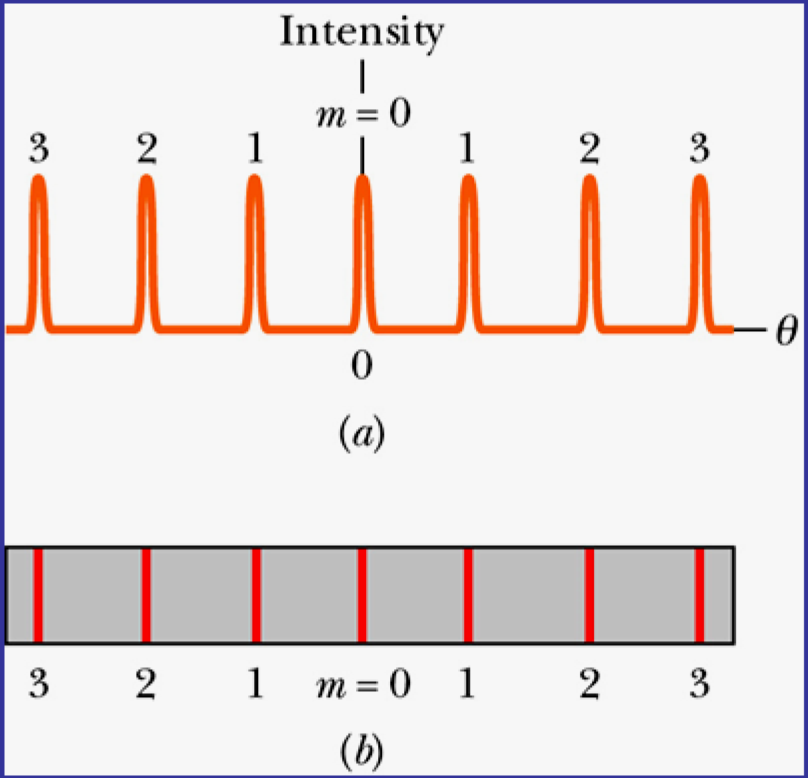
$$\beta = \frac{\pi d}{\lambda}\sin\theta$$

$$\alpha = \frac{\pi a}{\lambda}\sin\theta$$

Diffraction grating



System of N slits (rulings, grooves)

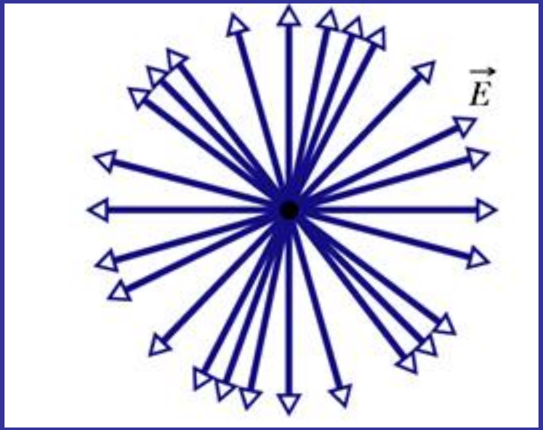


Maxima – diffraction lines appear when $d \sin\theta = m\lambda$

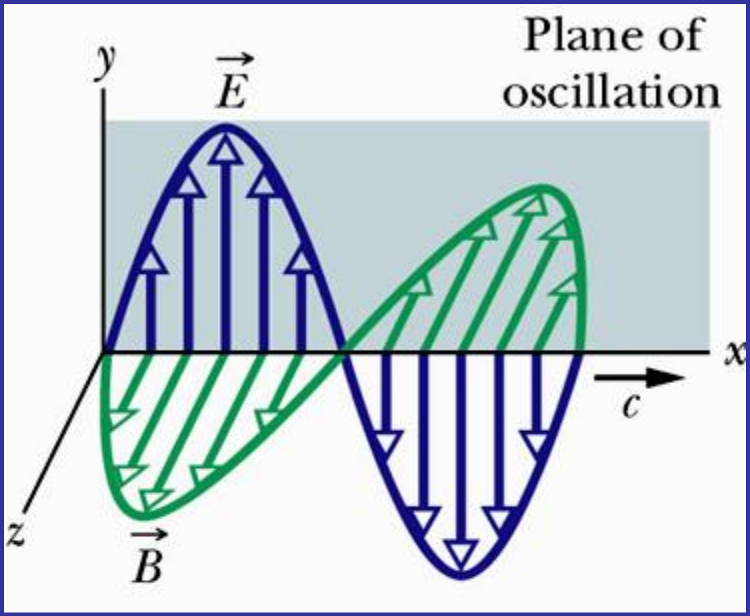
d - grating spacing

order numbers m=0,1,2,..

Polarization of the elelectromagnetic waves

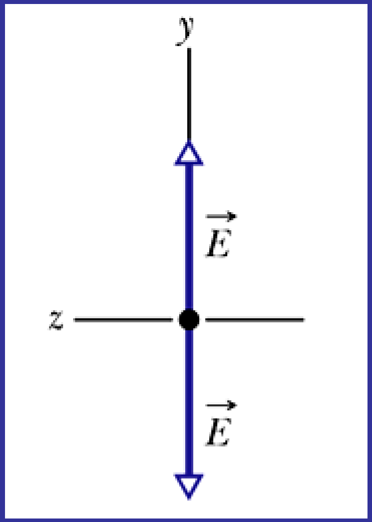


Unpolarized light



Plane-polarized

polarization



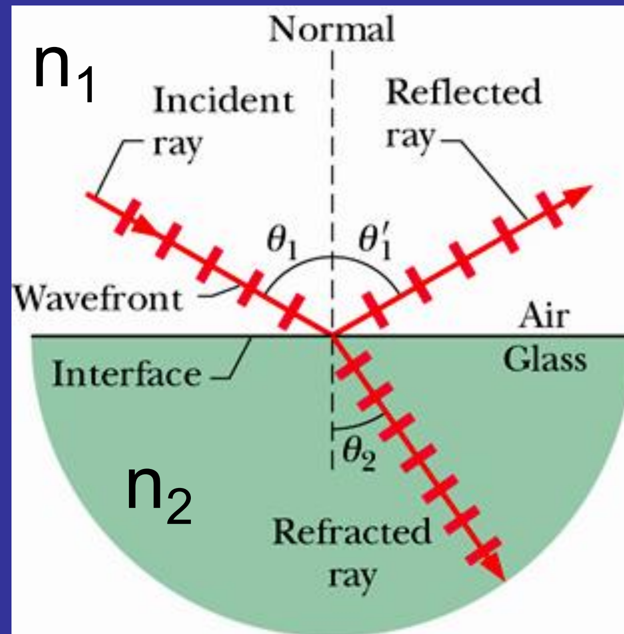
Linear polarization

Malus law

$$I(\theta) = I_o \cos^2 \theta$$

Reflection and refraction

Broken pencil – optical illusion



Law of reflection:

$$\theta_1 = \theta_1'$$

Snell's law = law of refraction

$$n_2 \sin \theta_2 = n_1 \sin \theta_1$$

Speed v of light depends on the medium in which it propages, different media, different refractive index n=c/v

Fermat rule

1679

Light travelling between two points in space choses a path which requires the minimum or maximum time (usually minimum) as compared with other possible paths in the vicinity.

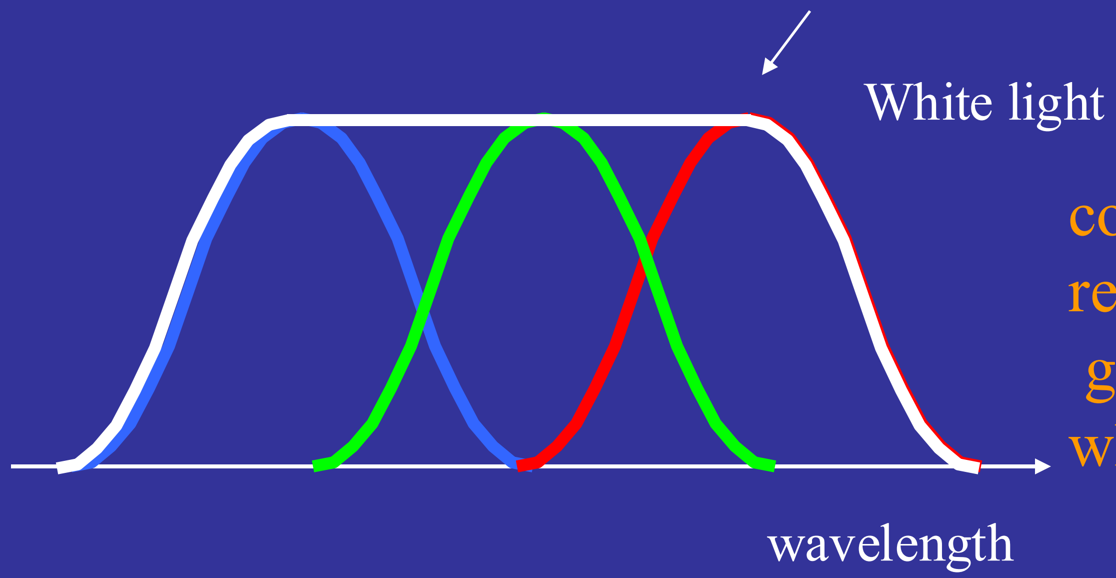
$$t = \int \frac{ds}{v} \Rightarrow t = \frac{1}{c}\int nds = \frac{optical\ \ path}{c}$$

Minimalization of time means minimalization of the optical path-length.

Fermat rule explains rectilinear propagation of light in a uniform medium, laws of reflection and refraction can be derived from this rule.

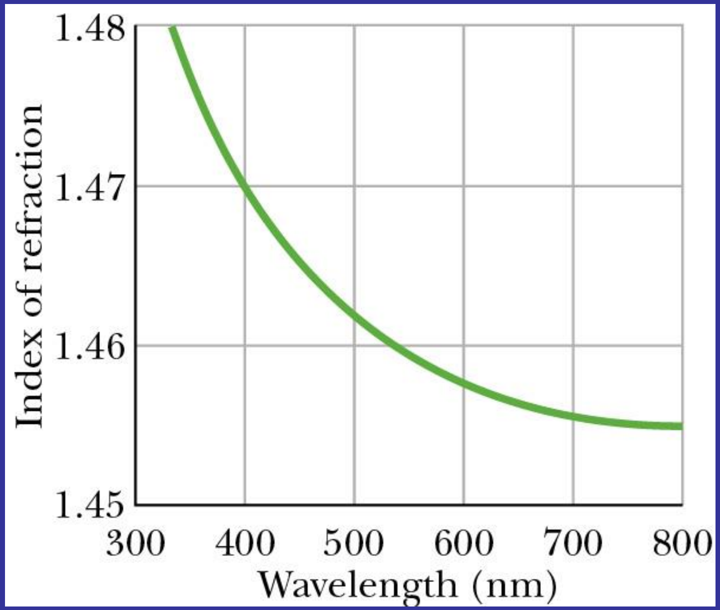
White light

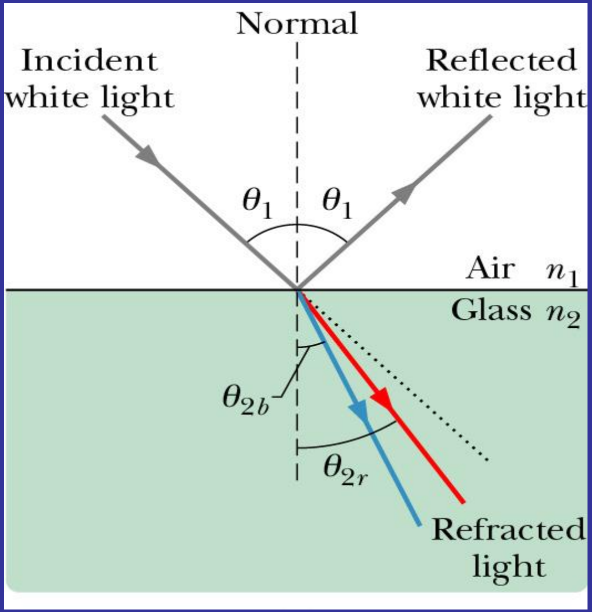
White light is an ideal combination of colors



color: blue, green and red mixed together give an impression of white light

Chromatic dispersion

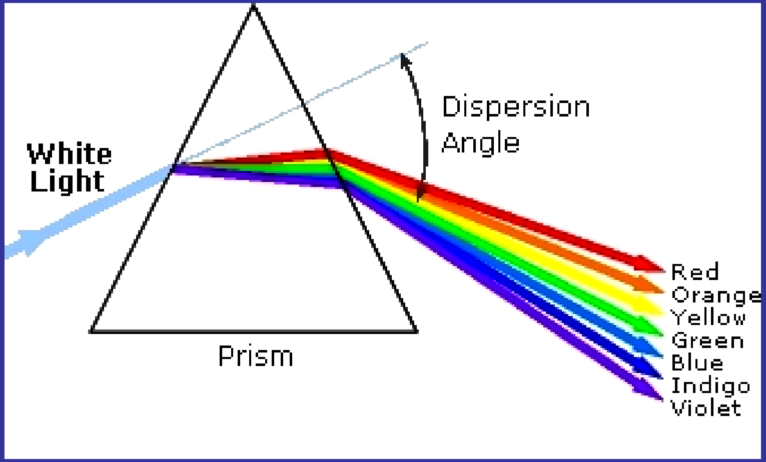




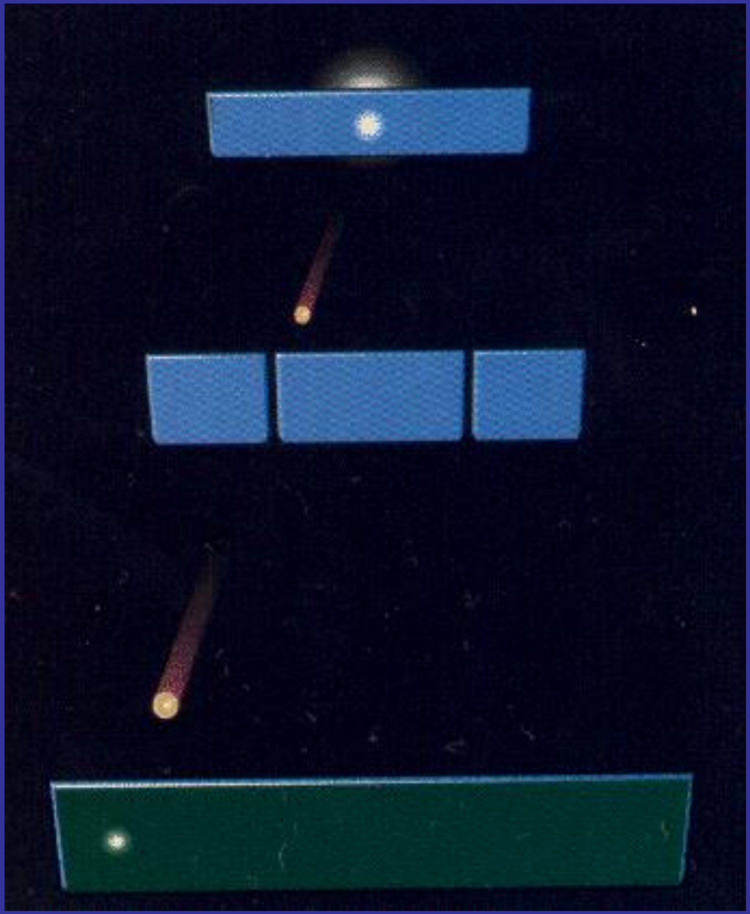
Monochromatic light of a given wavelength can be generated by means of :

dispersion n(λ) – prism

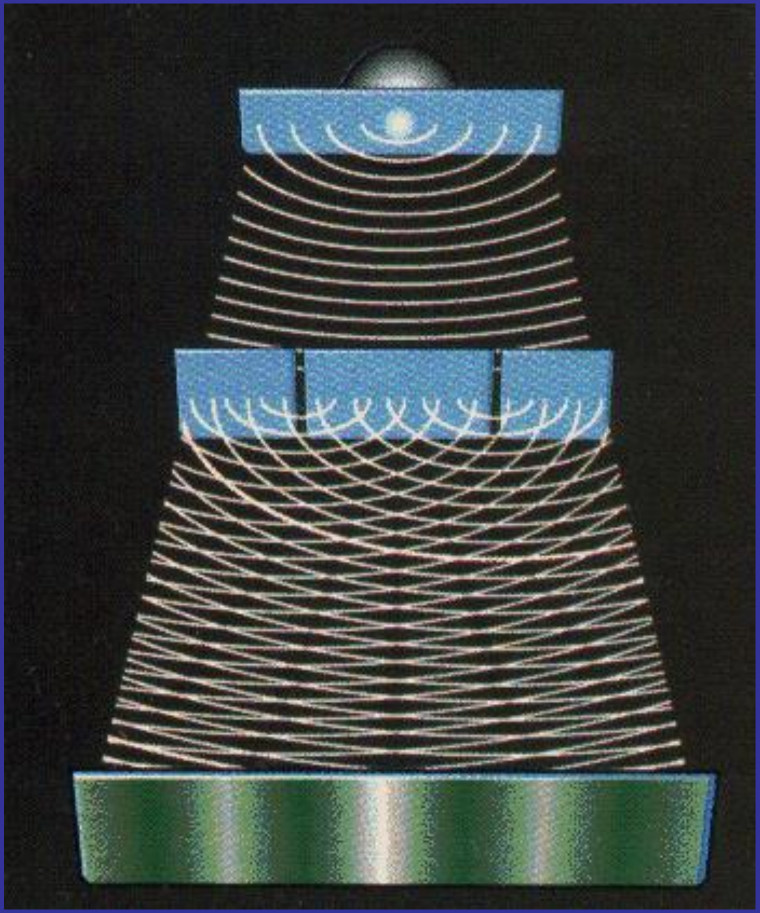
diffraction θ(λ) – diffraction grating



Nature of light



Light as a particle?



Light as a wave?

Particle-wave dualism:

In some experiments the wave character of light is demonstrated (diffraction, interference, polarization) while certain phenomena (photoelectric effect, Compton effect) can be explained based on a model assuming the existence of a quantum of the electromagnetic radiation – photon of energy $E=h\nu$ (h-Planck's constant)

Photon is a particle with zero rest mass.

Is electron a wave or a particle? Whether the waves of matter exist?

De Broglie’ hypothesis gives a positive answer to the last question.

Wavelength of wave associated with the particle

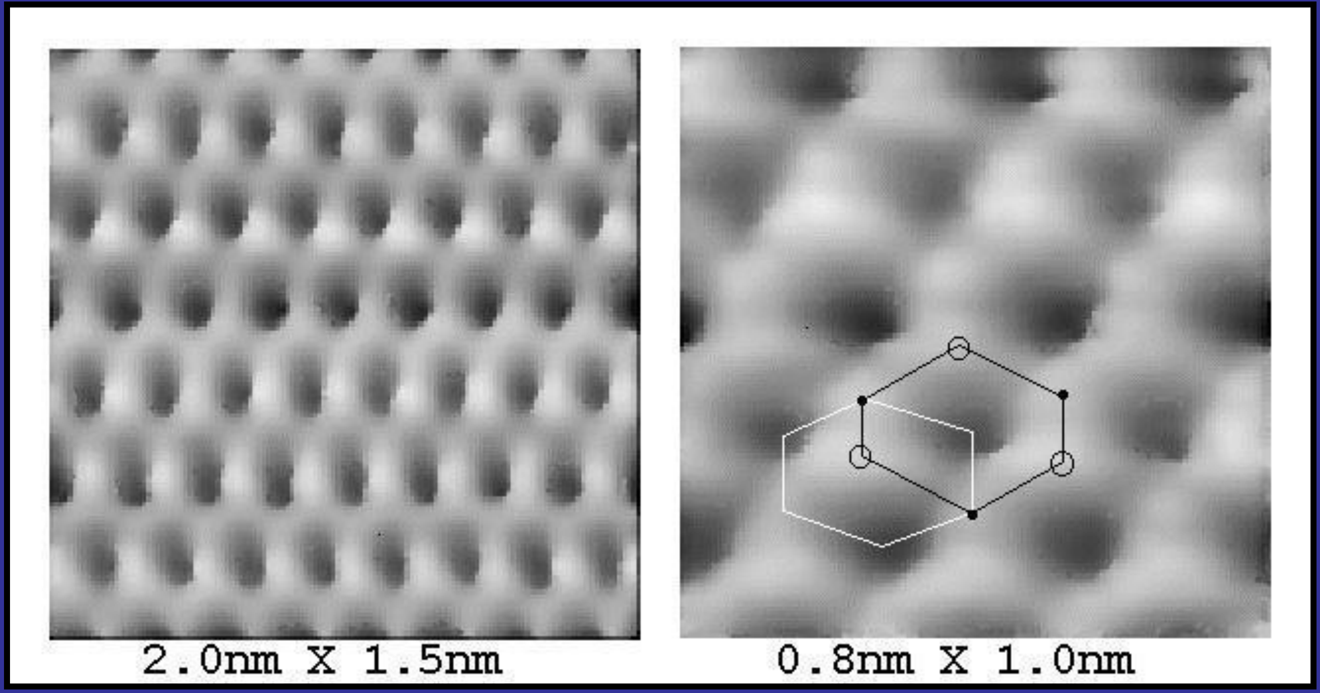
$$\lambda = \frac{h}{p}$$

Planck’s constant

Momentum of a particle

Diffraction of the electron waves really exists– transmission electron microscopy TEM

STM (Scanning Tunneling Microscope)



Resolution at an atomic scale